

NJALA UNIVERSITY



COLLEGE OF MEDICAL SCIENCES - BO CAMPUS School of Nursing

SECOND YEAR - 2024/2025 Academic Year

LECTURE NOTE

MODULE: HUMAN ANATOMY - BIOL- 212

{Endocrine, Nervous, Reproductive, Respiratory and Urinary Systems}

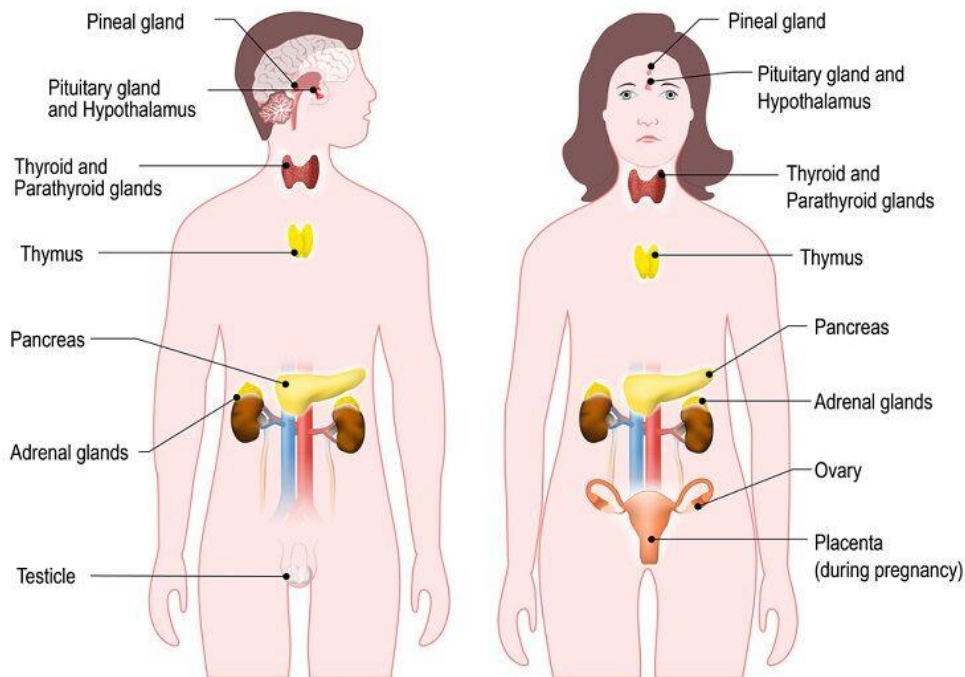
PROGRAMME: BACHELOR OF SCIENCE WITH HONORS IN NURSING

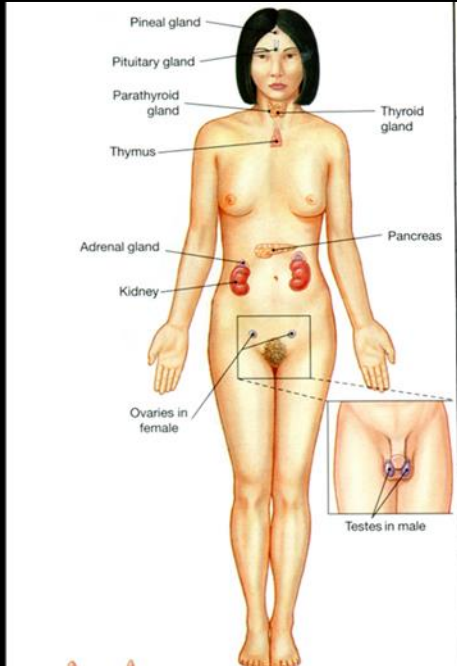
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ENDOCRINE SYSTEM



Organ/Component	Primary Functions
 Pineal gland	May control timing of reproduction and set day-night rhythms
Pituitary Gland	Controls other endocrine glands; regulates growth and fluid balance
Thyroid Gland	Controls tissue metabolic rate; regulates calcium levels
Parathyroid Glands	Regulate calcium levels (with thyroid)
Thymus	Controls maturation of lymphocytes
Adrenal Glands	Adjust water balance, tissue metabolism, cardiovascular and respiratory activity
Kidneys	Control red blood cell production and assist in calcium regulation
Pancreas	Regulates blood glucose levels
Gonads	
Testes	Support male sexual characteristics and reproductive functions (see part k)
Ovaries	Support female sexual characteristics and reproductive functions (see part l)

Introduction

Although often overshadowed by the brain and nerves, the Endocrine system is also involved in the information business. Hormones carry essential messages that have far-reaching effects. They control body processes at every level, from the energy uptake of a single cell to the whole body's rate of growth and development.

There are 50 hormones, which are the body's chemical messengers, and they are made by 12 different Endocrine glands. These glands have no ducts but secrete their hormones directly into the blood, which means they reach every cell in the body. Hormones affect certain target tissues or organs and regulate their activities.

The Endocrine system sends hormones through the Circulatory system to control and coordinate body functions in much the same way as the nervous system uses tiny electrical signals.

- The Endocrine system and the Nervous system work together to integrate into the brain and complement each other, but they tend to work at different speeds.
- Nerves respond within split-seconds, but their action soon fades
- Some hormones have longer-lasting effects and act over hours, weeks, and years.

The endocrine system

- Consists of many ductless glands which secrete hormones directly into the bloodstream. These hormones affect processes like metabolism, sex drive etc.
- Interacts closely with the nervous system.
- **Endocrinology**- the study of hormones and endocrine glands.

Hormones

- Chemical substances secreted in a small amount from the endocrine gland directly to the bloodstream in response to stimulus to cause physiological responses at the target tissues.
- need a specific receptor on a specific cell to do their job

Hormones affect several processes in the body, such as:

- ✚ Metabolism
- ✚ Sleep
- ✚ Sexual development and function
- ✚ Blood pressure
- ✚ Temperature
- ✚ Development and growth

Basic categories of hormones

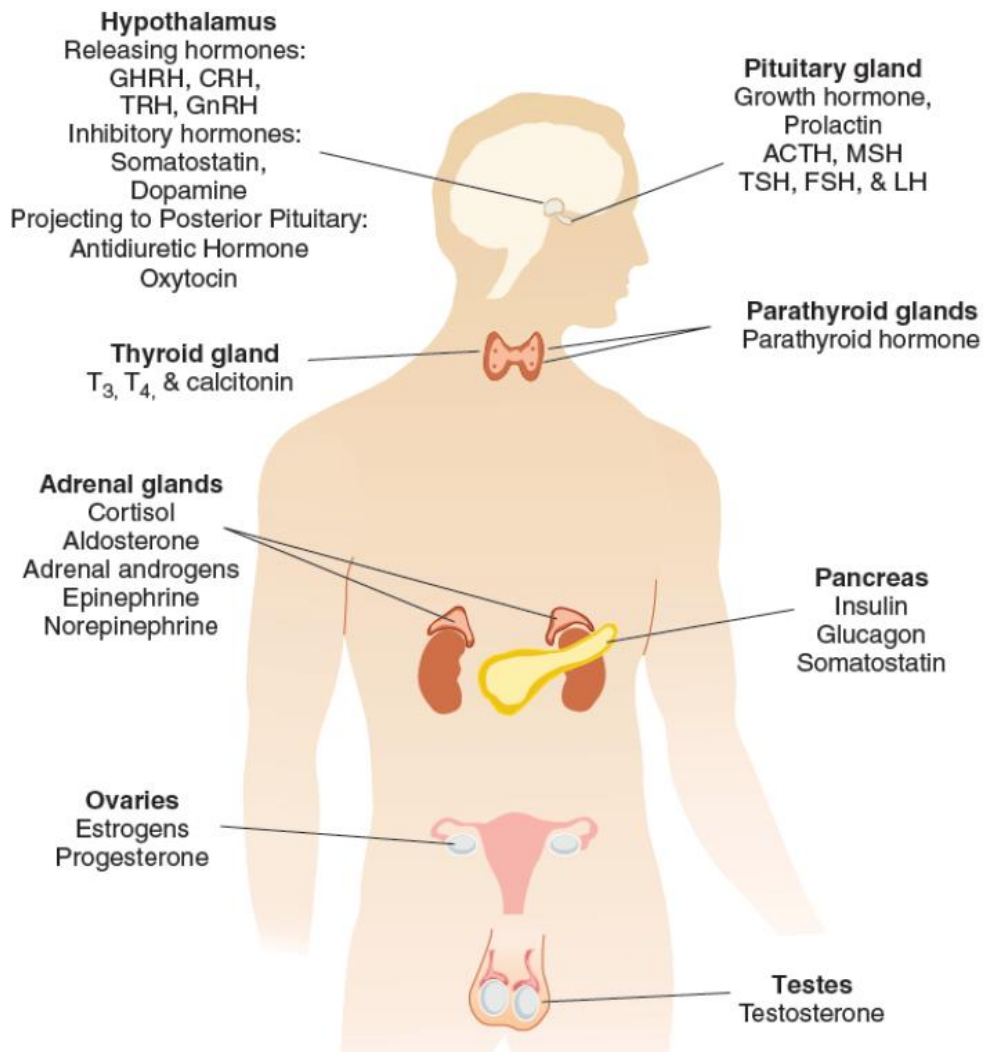
- Amino acid-based: modified amino acids (or *amines*), peptides (short chains of amino acids), and proteins (long chains of amino acids)
- Steroids: lipid molecules derived from cholesterol

The endocrine system is primarily controlled by the pituitary gland and the hypothalamus.

The function of the Endocrine System

The main function of the endocrine system is to regulate and monitor the levels of hormones circulating in the blood. This is because the endocrine system produces hormones that can regulate moods, growth, metabolism, development, and reproduction, which are released by what the body needs. One gland, the pituitary gland, is primarily responsible for sensing when hormone levels are increased and relays information to the other glands to stop producing the hormones. The pituitary gland can also sense when hormone levels are too low and instruct other glands to start producing more hormones.

Endocrine Organs and hormones

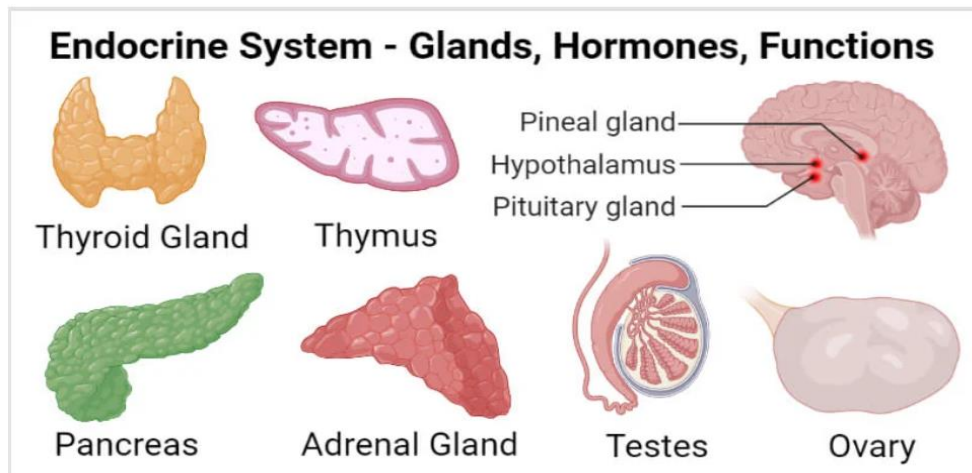


Purely endocrine organs

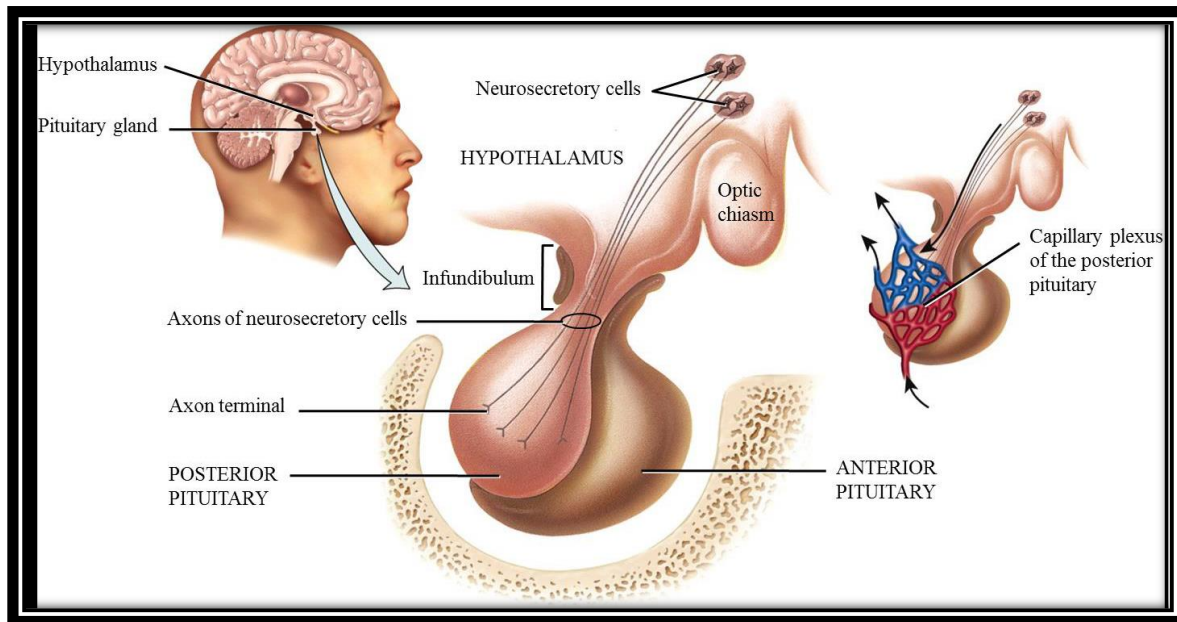
- Pituitary gland
- Pineal gland
- Thyroid gland
- Parathyroid glands
- Adrenal: 2 glands
 - ✓ Cortex
 - ✓ Medulla

Endocrine cells in other organs

- Pancreas
- Thymus
- Gonads
- Hypothalamus



HYPOTHALAMUS



- The hypothalamus is a small organ situated in the brain below the thalamus, which controls the secretion of the pituitary gland.
- The hypothalamus is attached to the pituitary gland by a small stalk called the infundibulum, and it is considered the connecting link between the endocrine system and the nervous system.
- The hypothalamus controls the endocrine system by controlling the pituitary gland.
 - Secretes releasing hormones to cause the pituitary to release hormones
 - Secretes inhibiting hormones to turn off the secretion of pituitary hormones
- The cells of the hypothalamus alone secrete about nine different hormones, out of which seven hormones are involved in the regulation of the pituitary gland.

General Functions of the Hypothalamus

1. Controls body temperature.
2. Controls the cardiovascular system.
3. Controls food intake & body weight.
4. Controls thirst & water balance.
5. Involved in sleep & wakefulness.
6. Involved in emotional reactions.
7. Involved in reactions to stress.
8. Controls ovarian secretions during the ovarian cycle.
9. Controls testicular secretions

Hormones secreted by the hypothalamus

All of these are released into the blood in the capillaries and travel immediately in portal veins to a second capillary bed in the anterior lobe of the pituitary, where they exert their effects. All of them are released in periodic spurts.

- Gonadotropin-releasing hormone (GnRH), Its secretion at the onset of puberty triggers sexual development, and from then on it is essential for normal sexual physiology in both males and females.
- Growth Hormone Releasing Hormone (GHRH), secrete growth hormone (GH)
- Thyrotropin Releasing Hormone (TRH), stimulates the release of TSH and PRL
- Corticotropin Releasing Hormone (CRH), acts on cells in the anterior lobe of the pituitary to release adrenocorticotrophic hormone (ACTH)
- Somatostatin, acts on the anterior lobe of the pituitary to inhibit the release of growth hormone (GH) and thyroid-stimulating hormone (TSH).
- Prolactin Inhibiting Factor (Dopamine)
- Peptide/Protein

The hypothalamus controls anterior pituitary hormone release

THE PITUITARY (Master Gland)

- The pituitary gland is a small pea-shaped organ measuring about 1-1.5 cm in diameter, which occurs in the hypophyseal fossa of the sphenoid bone in the skull.
- The pituitary gland is attached to the hypothalamus by the infundibulum (stalk)
- The pituitary gland can be differentiated into two anatomically and functionally separate parts, anterior pituitary and posterior pituitary.
 - Adenohypophysis (anterior lobe) accounts for about 70% of the total pituitary gland.
 - Neurohypophysis (posterior lobe)
- The anterior pituitary is supplied with a portal system that ensures the circulation of hormones produced by the gland.
- The secretion of the anterior pituitary is influenced by the releasing hormone produced by the hypothalamus.
- The posterior pituitary is composed of neural tissue, which is triggered by an action potential that originates in the cell body of the hypothalamus.
- **Secretes nine major hormones**

Anterior pituitary (adenohypophysis)

1. Luteinising Hormone (LH)

- ✓ Induces ovulation in females
- ✓ Induces testosterone in males

2. Follicle Stimulating Hormone (FSH)

- ✓ Present in males and females, affects both
- ✓ Stimulates maturation of sex cells

3. Growth Hormone (GH, somatotropin)

- ✓ Causes the body to grow

4. Prolactin (PRL)

- ✓ Stimulates lactation (milk production) in females
- ✓ Stimulate lacrimation (desire to cry)
- ✓ Decreased in adolescent males so it decreases the desire to cry

5. Thyroid Stimulating Hormone (TSH, thyrotropin)

- ✓ causes the thyroid gland to release thyroid hormone

6. Adrenocorticotrophic Hormone (ACTH, corticotropin)

- ✓ Acts on the adrenal cortex to stimulate the release of cortisol
- ✓ Helps people cope with stress

7. Melanocyte -Stimulating hormone (MSH)

- ✓ Darkens skin pigmentation
- ✓ Increases during pregnancy
- ✓ Also has effects on appetite and sexual arousal

Posterior pituitary (neurohypophysis)

8. ADH (antidiuretic hormone), or vasopressin

- ✓ Signals kidneys to increase water reabsorption

9. Oxytocin

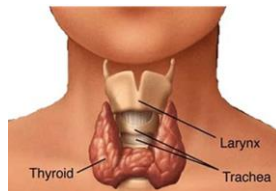
- ✓ It also stimulates the ejection of milk from the lactating breast.

THE PINEAL GLAND

The pineal gland is small, and pinecone shaped, (which is how it got its name) and located at the back of the diencephalon region in the brain.

- It secretes the hormone **Melatonin**
- At night, in the absence of light, the pineal gland secretes the hormone melatonin.
- Melatonin regulates the body's sleep patterns in both circadian (daily) and seasonal patterns.
- In the morning, when light hits the eye, photoreceptors in the retina send signals to the pineal gland, which then decreases melatonin production, and we wake up.

THYROID GLAND



The thyroid gland sits in the throat region, just below the larynx, served by large arteries with many branches and a dense network of capillaries. The hormones it secretes, travel in the bloodstream throughout the body to:

- Increase metabolism
- Regulate glucose use
- Protein synthesis
- Nervous system development.
- It also releases Calcitonin, which helps maintain blood calcium homeostasis by causing calcium to be removed from the blood and deposited into bones when blood (calcium) levels are too high.

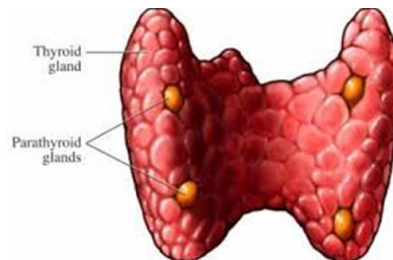
Produces two hormones

Thyroid hormone: tyrosine-based with 3 or 4 iodine molecules

- T4 (thyroxine) - 93% and T3 – 7% (Triiodothyronine and tetraiodothyronine)
- Calcitonin involved with calcium and phosphorus metabolism

Iodine required from dietary intake

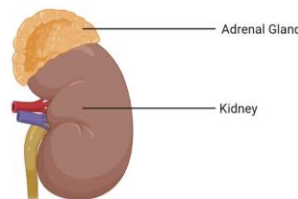
THE PARATHYROID GLANDS



On the posterior (back) surface of the thyroid sit much smaller, separate glands: the parathyroids. Typically, there are four parathyroid glands, a superior and inferior pair on the left and right sides of the thyroid.

- They secrete parathyroid hormone (PTH), which stimulates bones to release calcium into the blood when blood (calcium) levels are low.
- PTH also causes the kidneys to reduce calcium secretion into urine to further elevate calcium levels in the blood.
- Together, calcitonin and PTH act in complementary ways to maintain blood calcium homeostasis, which is one of the most tightly controlled physiological parameters in the body.
- Iodine is an element that's required for the thyroid gland to produce thyroid hormones.

ADRENAL (SUPRARENAL) GLANDS



Adrenal glands occur at the upper surface of each kidney which can be further divided into two distinct parts with different structures and functions.

- The adrenal cortex secretes three hormone types: **mineralocorticoids, glucocorticoids and androgens**
- The adrenal medulla secretes two major hormones: **epinephrine (adrenaline), 80%, and norepinephrine (noradrenaline 20%)**. Also, secret Dopamine

Glucocorticoids

- *Cortisol* (hydrocortisone), *corticosterone* and *cortisone* are the main glucocorticoids. They are essential for life, regulating metabolism and responses to stress. Secretion is stimulated by ACTH from the anterior pituitary and by stress.
- Glucocorticoids have widespread effects, and these include:
 - i. *gluconeogenesis* (formation of new sugar from, for example, protein) and hyperglycaemia (raised blood glucose level)
 - ii. *lipolysis* (breakdown of triglycerides into fatty acids and glycerol for energy production)
 - iii. stimulating the breakdown of protein, releasing amino acids, which can be used for the synthesis of other proteins, e.g. enzymes, or energy (ATP) production.
 - iv. promoting absorption of sodium and water from renal tubules (a weak mineralocorticoid effect).

Mineralocorticoids (aldosterone)

- *Aldosterone* is the main mineralocorticoid. Its functions are associated with the maintenance of water and electrolyte balance in the body. It stimulates the reabsorption of sodium (Na^+) by the renal tubules and excretion of potassium (K^+) in the urine. Sodium reabsorption is also accompanied by retention of water and therefore aldosterone is involved in the regulation of blood volume and blood pressure too.
- The blood potassium level regulates the amount of aldosterone produced by the adrenal cortex. When the blood potassium level rises, more aldosterone is secreted. Low blood potassium has the opposite effect.

Angiotensin also stimulates the release of aldosterone.

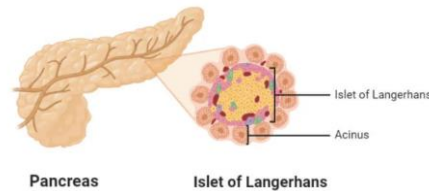
Androgens

- Androgens are also secreted by the adrenal cortex in males, but the concentration of those androgens is so low that their effects are almost insignificant.
- In females, however, androgens are responsible for sex drive and can also be converted into estrogens by other tissues.
- Adrenal androgens stimulate the growth of auxiliary and pubic hair in males and females.
- Androgens and Oestrogens
 - ✓ Male sex hormones (Androgens)
 - ✓ Female sex hormones (oestrogen)

Adrenal medulla

- Part of the autonomic nervous system
 - **Secrete epinephrine and norepinephrine**
 - Amine hormones
 - Fight, flight, and fright which leads to
 - Increase heart rate
 - Incr. BP
 - Send blood to essential organs
 - Incr. metabolism
 - Dilate pupil

PANCREAS



- The endocrine part of the pancreas is the cluster of cells, called pancreatic islets or Islet of Langerhans.
- These occur in the form of a cluster of cells scattered throughout the gland. The hormones produced by the cells are released directly into the bloodstream.
- The pancreatic islets consist of three different types of cells: *α cells, β cells, and δ cells*. The three different cells secrete three distinct hormones. A minor fourth type of cell also occurs in the pancreas called the PP cells.

Four Types of Secretory Cells

Alpha (A) cells, which secrete glucagons

Beta (B) cells, which secrete insulin

Delta (D) cells, which secrete somatostatin

PP (F) cells, which secrete pancreatic polypeptide

Glucagon

- glucagon increases blood glucose levels by stimulating, e.g. conversion of glycogen to glucose in the liver and skeletal muscles (glycogenolysis)
- gluconeogenesis. Secretion of glucagon is stimulated by a low blood glucose level and exercise and decreased by somatostatin and insulin.

Insulin

- The main function of insulin is to lower blood levels of absorbed nutrients when they rise above normal.
- When these nutrients, especially glucose, are more than immediate needs insulin promotes storage by:
 - ✓ acting on cell membranes and stimulating the uptake and use of glucose by muscle and connective tissue cells

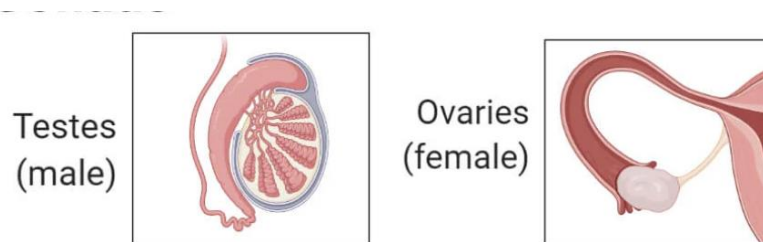
- ✓ increasing conversion of glucose to glycogen (glycogenesis), especially in the liver and skeletal muscles
- ✓ accelerating uptake of amino acids by cells, and the synthesis of protein
- ✓ promoting synthesis of fatty acids and storage of fat in adipose tissue (lipogenesis)
- ✓ decreasing glycogenolysis
- ✓ preventing the breakdown of protein and fat, and *gluconeogenesis* (formation of new sugar from, e.g. protein)

Somatostatin (GHRIH)

- The effect of this hormone, also produced by the hypothalamus, is to inhibit the secretion of both insulin and glucagon.

THE GONADS (testes and ovaries)

main source of the steroid sex hormones



Testes

- Interstitial cells secrete androgens
- Primary androgen is **testosterone**
- Testosterone is produced by the testes, which stimulates the maturation of sex organs, their development, and the appearance of secondary sex characters.
- The release of testosterone is maintained by the release of gonadotrophin-releasing hormone produced by the hypothalamus.
- Helps promote sperm formation

Ovaries

The hormones produced by the ovaries are: **Estrogen and Progesterone**

Estrogen

- **Estrogen** is produced by the cells in the ovaries, which are essential for the maturation of the reproductive organs and the appearance of secondary sex characters in females at puberty.
- Estrogens are analogous to male steroid testosterone. The level of estrogen in the body increases during puberty to promote oogenesis and follicle growth.
- The activity of estrogen results in the proper functioning of different reproductive parts like the uterine tubes, uterus, and vagina.

Progesterone

- Progesterone is produced by the corpus luteum during the menstrual cycle, which regulates the cycles and activates changes in the cervical mucosa.
- Most of the effects of the hormone are observed during pregnancy as it inhibits uterine mobility. The level of progesterone increases more than that of estrogen during pregnancy.

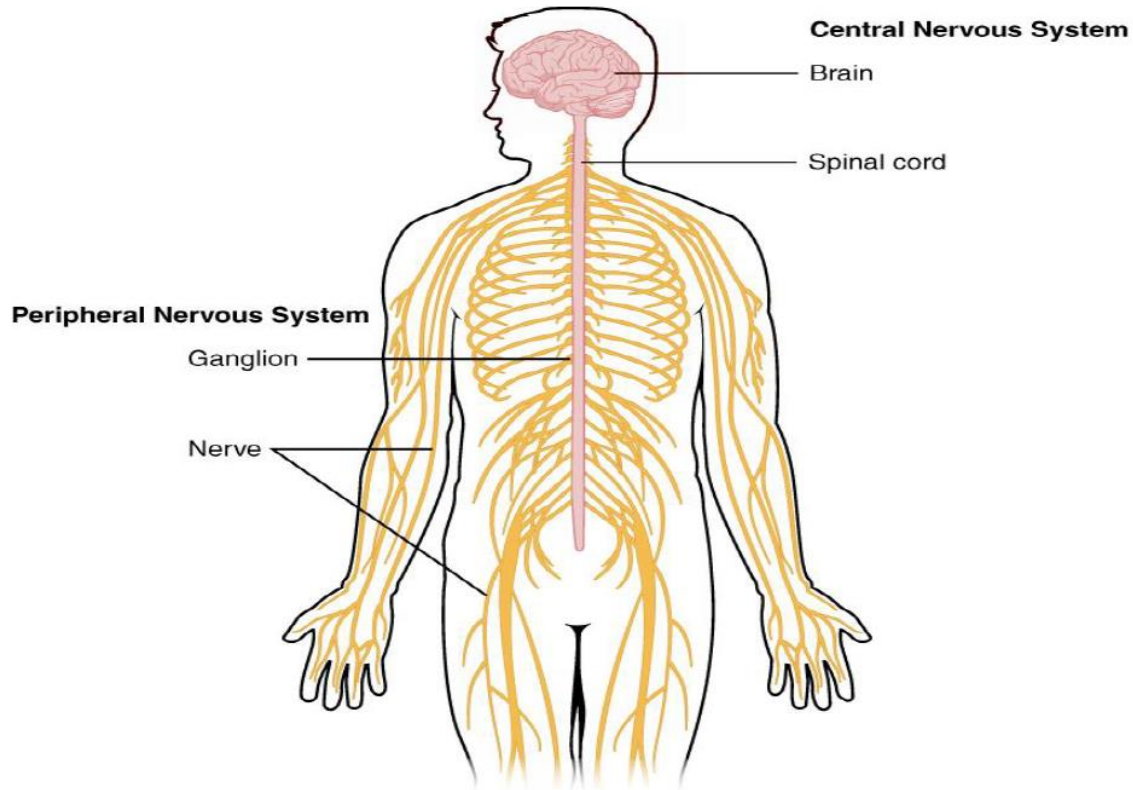
Summary

Endocrine gland	Hormone(s) secreted	Hormone function
Adrenal glands	Aldosterone	Regulates salt, water balance, and blood pressure
	Corticosteroid	Controls key functions in the body; acts as an anti-inflammatory; maintains blood sugar levels, blood pressure, and muscle strength; regulates salt and water balance
	Epinephrine	Increases heart rate, oxygen intake, and blood flow
	Norepinephrine	Maintains blood pressure
	Dehydroepiandrosterone (DHEA)	Used in the production of androgens (male sex hormones) and estrogens (female sex hormones) Has effects on bone, mood, and the immune system
Pituitary gland	Antidiuretic hormone (vasopressin)	Affects water retention in kidneys; controls blood pressure
	Adrenocorticotrophic hormone (ACTH)	Controls the production of sex hormones (estrogen in women and testosterone in men) and the production of eggs in women and sperm in men.
	Growth hormone (GH)	Affects growth and development; stimulates protein production; affects fat distribution
	Luteinizing Hormone (LH) & Follicle-stimulating hormone (FSH)	Controls the production of sex hormones (estrogen in women and testosterone in men) and the production of eggs in women and sperm in men
	Oxytocin	Stimulates contraction of the uterus and milk ducts in the breast
	Prolactin	Initiates and maintains milk production in breasts; impacts sex hormone levels
	Thyroid-stimulating hormone (TSH)	Stimulates the production and secretion of thyroid hormones
	Melatonin	Releases melatonin during night hours to help with sleep
Hypothalamus	Growth hormone-releasing hormone (GHRH)	Regulates growth hormone release in the pituitary gland
	Thyrotropin-releasing hormone (TRH)	Regulates thyroid-stimulating hormone release in the pituitary gland
	Gonadotropin-releasing hormone (GnRH)	Regulates LH/FSH production in the pituitary gland

	Corticotropin-releasing hormone (CRH)	Regulates adrenocorticotropin release in the pituitary gland
Pancreas	Glucagon	Raises blood sugar levels
	Insulin	Lowers blood sugar levels; stimulates metabolism of glucose, protein, and fat
Thyroid gland	Thyroid hormone	Controls metabolism; also affects growth, maturation, nervous system activity, and metabolism
Parathyroid glands	Parathyroid hormone (PTH)	Most important regulator of blood calcium levels
Ovaries	Estrogen	works to regulate the menstrual cycle, maintain pregnancy, and develop female sex characteristics; aids in sperm production
	Progesterone	Stimulates the lining of the uterus for fertilization; prepares the breasts for milk production
Testes (testicles)	Testosterone	Develop and maintain male sexual characteristics and maturation, contributes to sex drive and body density in males.

Nervous system	Endocrine system
Nervous system regulates the activities of muscles and glands via electrical impulses transported through neurons	Endocrine System regulates the body's metabolic activity via hormones that are transported in the blood
Reflexes of the nervous system are more specific because nerves regulate specific target cells	Hormones can control multiple organs and systems
Nervous system uses both electrical (action potentials) and chemical (neurotransmitters) signals	Endocrine system uses only chemical signals (hormones)
Neural control is fast; its effects are short-lived	Hormonal control is comparatively slower; generally its effects are prolonged
Stimulus intensity in the nervous system is represented by the frequency and amplitude of action potentials.	Stimulus intensity in the endocrine system is determined by the amount and duration of hormone(s) released.

THE NERVOUS SYSTEM



Introduction

The nervous system is the most complex body system. Constantly alive with electricity, the nervous system is the body's prime communication and coordination network. It consists of:

- ✦ Nerve cells (neurons)
- ✦ Glial cells
- ✦ Nerve fibres

Functions

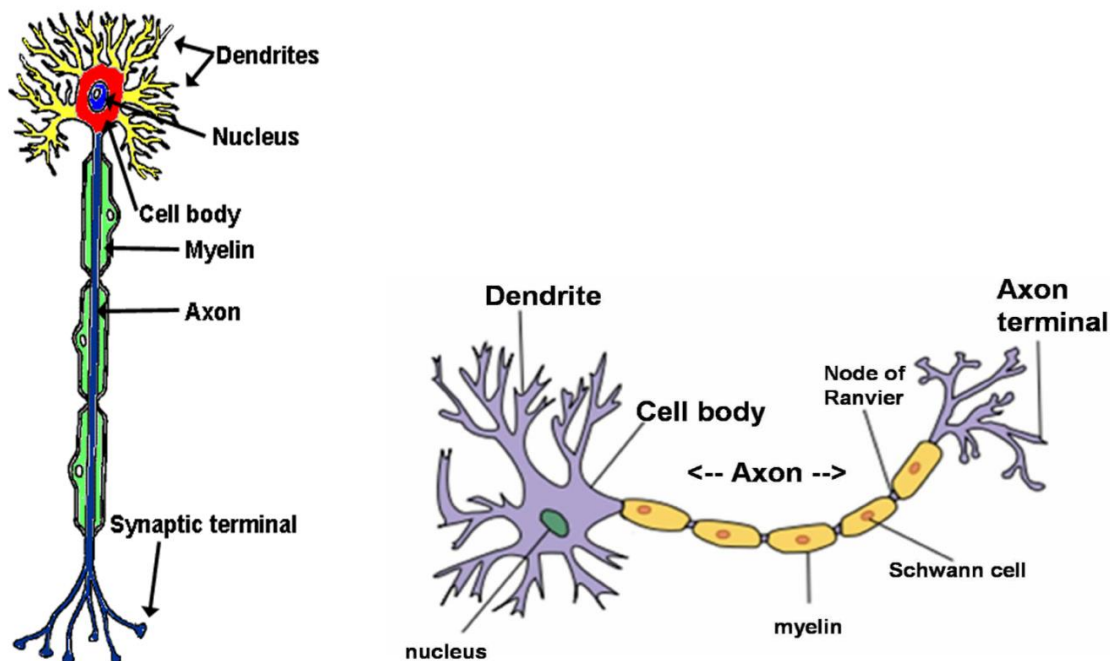
- **Monitoring changes** - Much like a sentry, it uses its millions of sensory receptors to monitor changes occurring both inside and outside the body; these changes are called stimuli, and the gathered information is called sensory input.
- **Interpretation of sensory input** - It processes and interprets the sensory input and decides what should be done at each moment, a process called integration.
- **Effects responses** - It then affects a response by activating muscles or glands (effectors) via motor output.
- **Mental activity** - The brain is the centre of mental activity, including consciousness, thinking, and memory.
- **Homeostasis** - This function depends on the ability of the nervous system to detect, interpret, and respond to changes in internal and external conditions. It can help stimulate

or inhibit the activities of other systems (the endocrine which is the second most regulatory system) to help maintain a constant internal environment.

Parts of the Nervous System (Nerves, Brain, and Spinal Cord)

Nerve Cells (Neurons)

- Neurons (Basic functioning units of the nervous system)
- Made up of a cell body and branches called dendrites and axons.
- More than 100 billion neurons form a network for communication.
- Each neuron has an average of at least a thousand interconnections with other neurons, forming a very complex system for communication.
- Neurons are responsible for the reception, transmission, and processing of stimuli.
- A message carried by a neuron is called an impulse.



- **Dendrites** - carry impulses towards the cell
- **Axon** - carry impulses away from the cell
- **Myelin sheath** - dense lipid layer which insulates the axon
- **Synaptic terminal**
 - ✦ Epinephrine
 - ✦ Norepinephrine
 - ✦ Acetylcholine

Types of Neurons.

1. According to their functions

- Sensory
- Motor
- Interneuron

Sensory Neurons

- ✚ Receive information.
- ✚ Send impulses to the brain or spinal cord

Motor Neurons

- ✚ Conduct impulses from the brain or spinal cord to muscles or glands throughout your body.

Interneurons

- ✚ Send impulses from sensory neurons to motor neurons.

Properties of Neurons

1. **Irritability:** the ability to initiate nerve impulses in response to stimuli from; outside and inside the body
2. **Conductivity:** the ability to transmit an impulse.

The nerve impulse (action potential)

- An impulse is initiated by stimulation of sensory nerve endings or by the passage of an impulse from another nerve.
- Transmission of the impulse, or action potential, is due to the movement of ions across the nerve cell membrane. In the resting state, the nerve cell membrane is *polarized* due to differences in the concentrations of ions across the plasma membrane.

2. Types of neurons according to the shape and size of their processes

- i. **Multipolar neurons** have more than 2 cell processes (one process being the axon and the other dendrites).
- ii. **Bipolar neurons** have one dendrite and one axon.
- iii. **Pseudounipolar neurons** have a single process that is close to the cell body and divides into two branches.

Glial Cells (neuroglia) / Supporting cells.

These cells have supporting and protective functions in the nervous tissue.

The neuroglia includes:

1. Astrocytes (the most abundant type of glial cells) have supporting functions and control the ionic and chemical environment of neurons.
2. Oligodendrocytes produce the myelin sheath for axons in the central nervous system.
3. Schwann cells produce myelin sheath in the peripheral nervous system.
4. Ependymal cells found in the central nervous system which facilitate the movement of cerebral spinal fluids.
5. Microglia have phagocytic functions.

Organization of the Nervous System

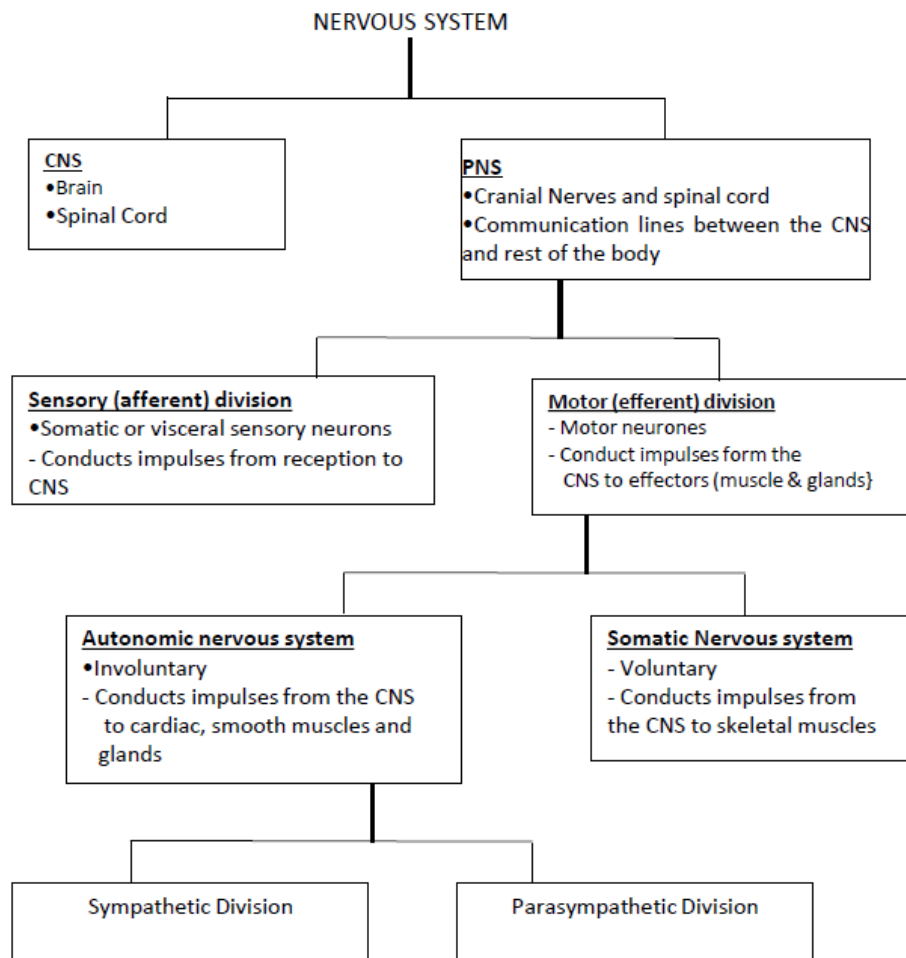
It's divided in terms of

1. Structures (structural classification)
2. Activities (functional classification).

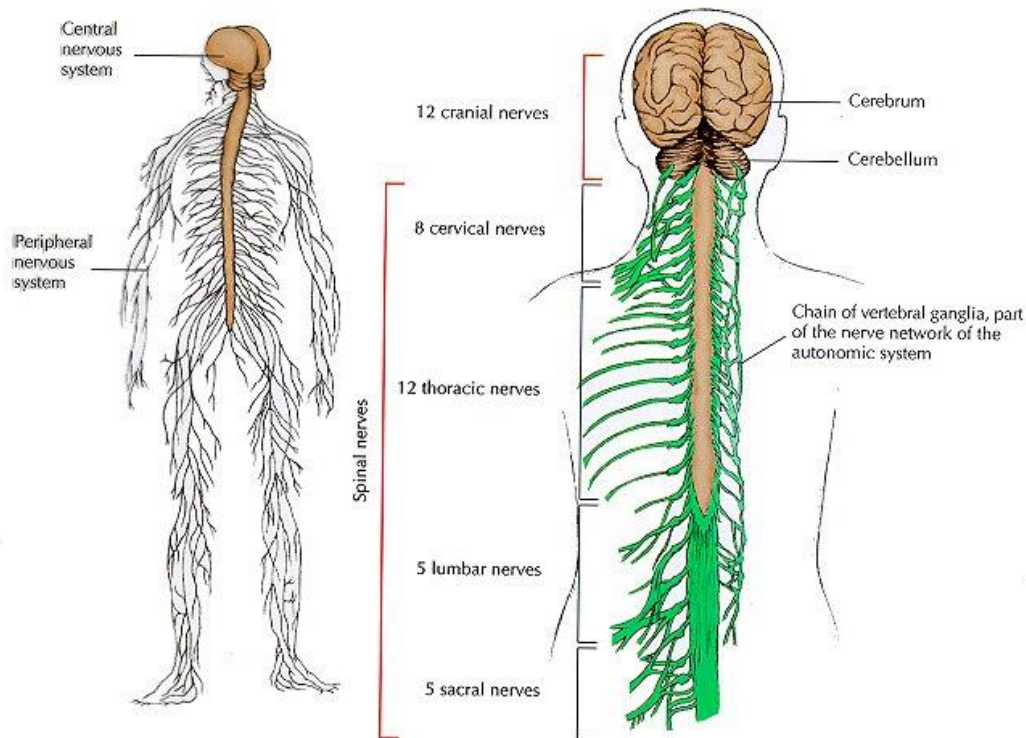
Structural Classification

Has two subdivisions

- The CNS
- The PNS.



The Central and Peripheral Nervous Systems



Central Nervous System (CNS)

- The brain and spinal cord, act as the integrating and command centres of the nervous system. It interprets incoming sensory information and dictates responses based on experience, reflexes, and current conditions.

Peripheral Nervous System (PNS)

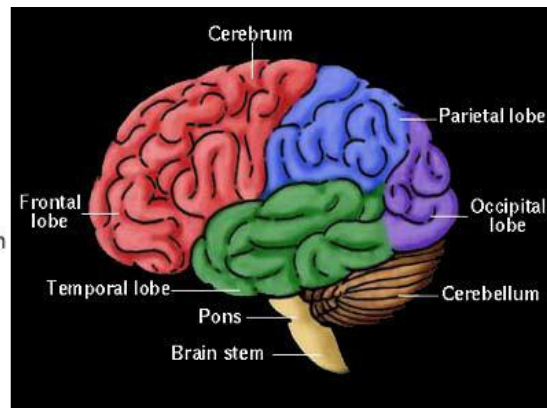
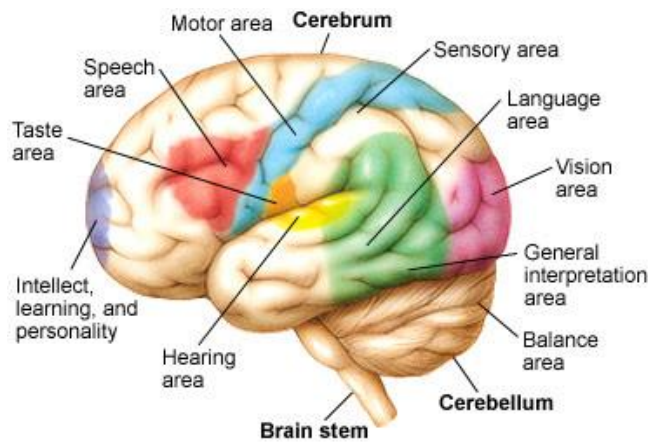
- The nervous system outside the CNS, consists mainly of the nerves that extend from the brain and spinal cord. (Cranial And Spinal Nerves)

Central Nervous System

- Brain
- Spinal cord

The Brain

- Coordinates body activities
- Made up of approximately 100 billion neurons.
- Divided into four major parts-
 - Diencephalon (thalamus, hypothalamus)
 - the cerebrum
 - the cerebellum
 - the brain stem (midbrain, pons, and medulla)



Cerebrum

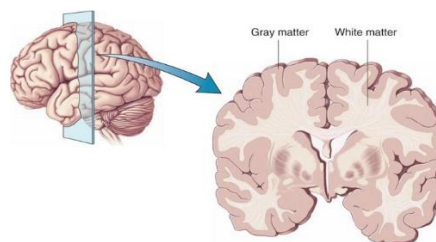
The largest division of the brain. It is divided into two hemispheres, each of which is divided into four lobes.



Functions

- Largest part of the brain
- Thinking
- Memory is stored.
- Movements are controlled.
- Impulses from the senses are interpreted.

Cerebral Cortex - The outermost layer of gray matter makes up the superficial aspect of the cerebrum.



Cerebral Features:

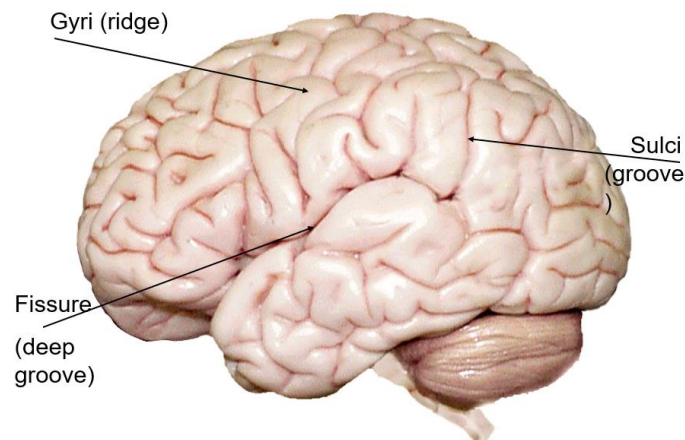
Gyri – Elevated ridges “winding” around the brain.

Sulci – Small grooves dividing the gyri.

- Central Sulcus – Divides the Frontal Lobe from the Parietal Lobe

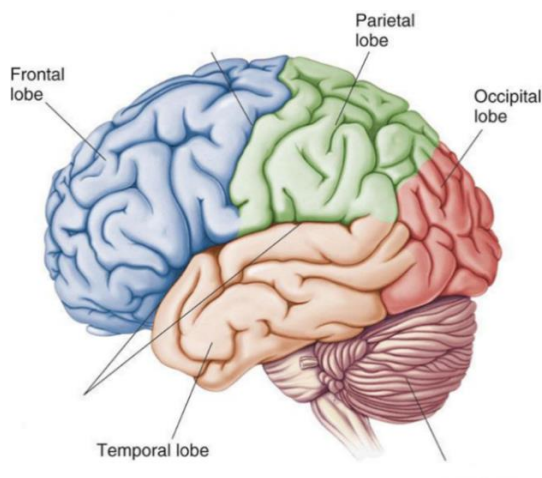
Fissures – Deep grooves, generally dividing large regions/lobes of the brain.

- **Longitudinal Fissure** – Divides the two Cerebral Hemispheres
- **Transverse Fissure** – Separates the Cerebrum from the Cerebellum
- **Sylvian/Lateral Fissure** – Divides the Temporal Lobe from the Frontal and Parietal Lobes



Lobes of the Brain (4)

- Frontal
- Parietal
- Occipital
- Temporal



Frontal Lobe

The Frontal Lobe of the brain is located deep in the Frontal Bone of the skull. It plays an integral role in the following functions/actions:

- Memory Formation
- Emotions
- Decision
- Making/Reasoning
- Personality

Parietal Lobe

The Parietal Lobe of the brain is located deep in the Parietal Bone of the skull.

It plays a major role in the following functions/actions:

- Senses and integrates sensation(s)
- Spatial awareness and perception (Proprioception - Awareness of body/ body parts in space and about each other)

Occipital Lobe

The Occipital Lobe of the Brain is located deep in the Occipital Bone of the Skull

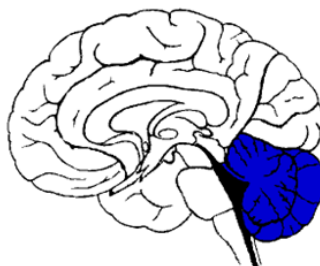
- Its primary function is the processing, integration, interpretation, etc. of VISION and visual stimuli.

Temporal Lobe

The Temporal Lobes are located on the sides of the brain, deep into the Temporal Bones of the skull. They play an integral role in the following functions:

- Hearing
- Organization/Comprehension of language
- Information Retrieval (Memory and Memory Formation)

Cerebellum



- The 2nd largest structure of the brain
- Located at the inferior or posterior aspect of the cranial cavity.
- Interprets stimuli from eyes, ears, and muscles.
- Controls voluntary muscle movements
- Maintains muscle tone.
- Helps maintain balance.

- Has two hemispheres and a central constricted area called the vermis.
- Cerebellar peduncles – support and communicate with another part of the brain.
- It has a cortex, gray matter, and a deeper layer of white matter.
- It also has gyri; it also has a distinctive branching pattern called arbour vitae- seen in a sagittal view.

Principal functions

- Impulse for voluntary muscular movement starts from the cerebral cortex and is coordinated by the cerebellum.
- Initiates impulse to selective motor units for maintaining postures and muscle tone.
- It adjusts impulses from proprioceptors within muscles, tendons, joints and special sense organs.

Brain Stem

- Connects brain to spinal cord.
- Made up of the midbrain, the pons,
 - Act as pathways connecting various parts of the brain.
- Medulla oblongata
 - controls involuntary actions

Midbrain

- ✦ The midbrain is the area of the brain situated around the cerebral aqueduct between the cerebrum above and the *pons* below.
- ✦ It consists of groups of cell bodies and nerve fibres (tracts) which connect the cerebrum with the lower parts of the brain and with the spinal cord.
- ✦ The cell bodies act as relay stations for the ascending and descending nerve fibres.

Pons

- ✦ The pons is situated in front of the cerebellum, below the midbrain and above the medulla oblongata.
- ✦ It consists mainly of nerve fibres which form a bridge between the two hemispheres of the cerebellum, and fibres passing between the higher levels of the brain and the spinal cord.
- ✦ There are groups of cells within the pons which act as relay stations and some of these are associated with the cranial nerves.
- ✦ The anatomical structure of the pons differs from that of the cerebrum in that the cell bodies (grey matter) lie deeply, and the nerve fibres are on the surface.

Medulla oblongata

- The medulla oblongata extends from the pons above and is continuous with the spinal cord below.
- It is about 2.5 cm long and it lies just within the cranium above the foramen magnum. Its anterior and posterior surfaces are marked by central fissures.
- The outer aspect is composed of *white matter* which passes between the brain and the spinal cord, and *grey matter* lies centrally. Some cells constitute relay stations for sensory nerves passing from the spinal cord to the cerebrum.

Diencephalon

- The diencephalon is located between the brainstem and the cerebrum.

Thalamus

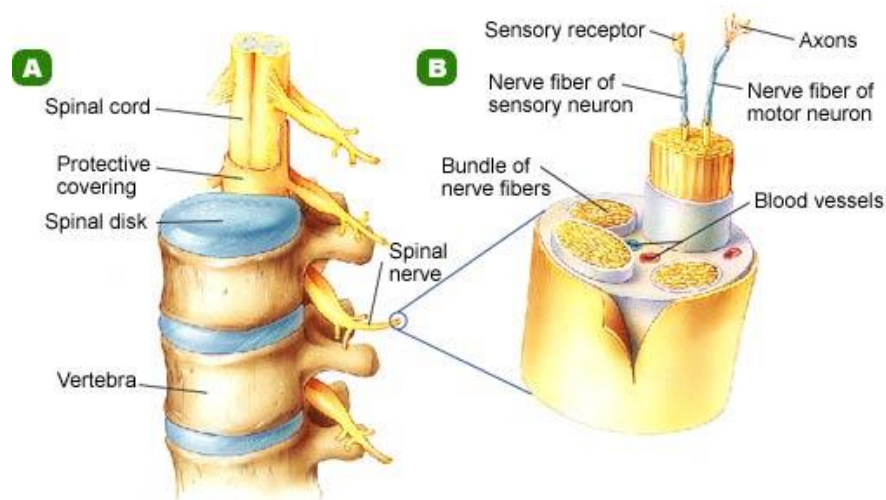
- ✦ The thalamus consists of two lobes connected by the interthalamic adhesion. The thalamus functions as an integration centre.
- ✦ Most sensory input synapses in the thalamus. Pain is registered in the thalamus.
- ✦ The thalamus also has some motor functions.

Hypothalamus

- ✦ The hypothalamus, the most inferior portion of the diencephalon, contains several nuclei and tracts.
- ✦ The mammillary bodies are reflex centres for olfaction.
- ✦ The hypothalamus regulates many endocrine functions (e.g., metabolism, reproduction, response to stress, and urine production). The pituitary gland attaches to the hypothalamus.
- ✦ The hypothalamus regulates body temperature, hunger, thirst, satiety, swallowing, and emotions.
 - The vital centres, consisting of groups of cells associated with autonomic reflex activity, lie in its deeper structure.
 - These are the:
 - ✓ cardiac centre
 - ✓ respiratory centre
 - ✓ vasomotor centre
 - ✓ reflex centres of vomiting, coughing, sneezing and
 - ✓ swallowing

The Spinal Cord

The cylindrical spinal cord is a glistening white continuation of the brain stem.



- Extension of the brain stem
- Bundles of neurons that carry impulses from all parts of the body to the brain and from the brain to all parts of your body.
- It is continuous above with the medulla oblongata and extends from the *upper border of the atlas* to the lower border of the 1st *lumbar vertebra*.
- The spinal cord is approximately 17 inches (42 cm) long.
- It is communication link between the brain and the PNS inferior to the head.
- The spinal cord extends from the foramen magnum to the level of the second lumbar vertebral column during development.
- It is composed of cervical, thoracic, lumbar, and sacral segments.
- It is named according to the portion of the vertebral column from which their nerves enter and exit.

The spinal cord gives rise to 31 pairs of spinal nerves, which exit vertebral column through the intervertebral foramina.

- Like the brain, the spinal cord is cushioned and protected by the meninges; meningeal coverings end at well beyond the end of the spinal cord in the vertebral canal.
- The collection of spinal nerves at the inferior end of the vertebral canal is called cauda equina because it looks so much like a horse's tail.

Protection of the central nervous system

It is protected by:

- skull and vertebral column
- Meninges
- Cerebrospinal fluid
- Blood-brain barrier

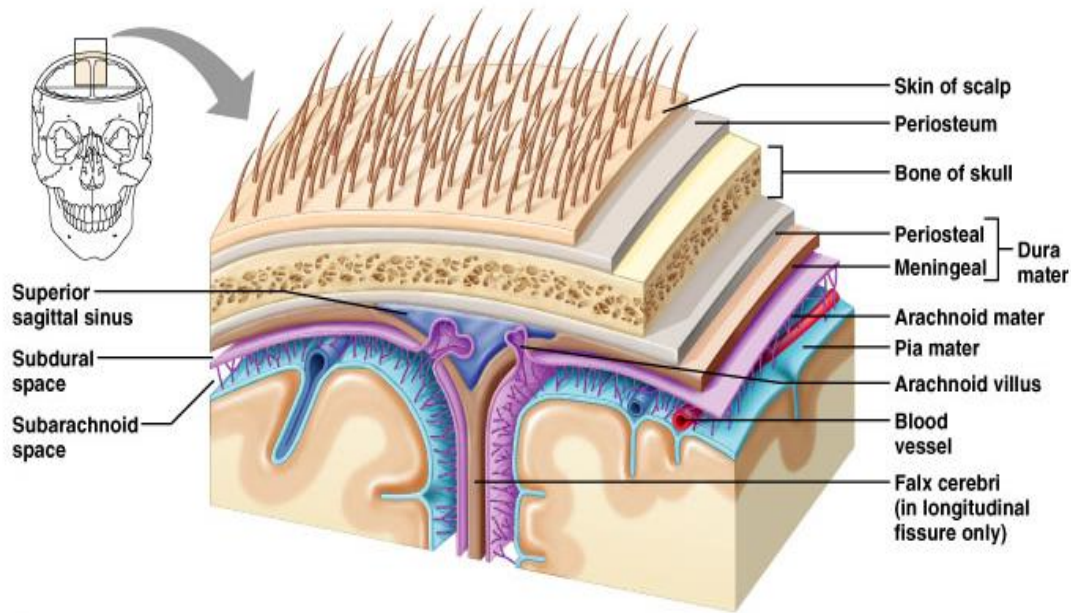
Cranial Meninges - 3-layer protective membrane

1. **Dura Mater** - Composed of two layers:

- a) Periosteal – outer layer, attaches to bone.
- b) Meningeal – inner layer, closer to the brain.

2. **Arachnoid Layer** - 'spider' web-like.

3. **Pia Mater** - delicate, follows convolutions.



(a)

Cranial Meningeal Spaces

Epidural space

- Potential space superior to the dura.

Subdural space

- Potential space between dura and arachnoid mater.

Subarachnoid space

- Between the arachnoid mater and the pia mater is the subarschnoid space which contains the cerebrospinal fluid
- Filled with CSF
- Contains the blood vessels supplying brain.

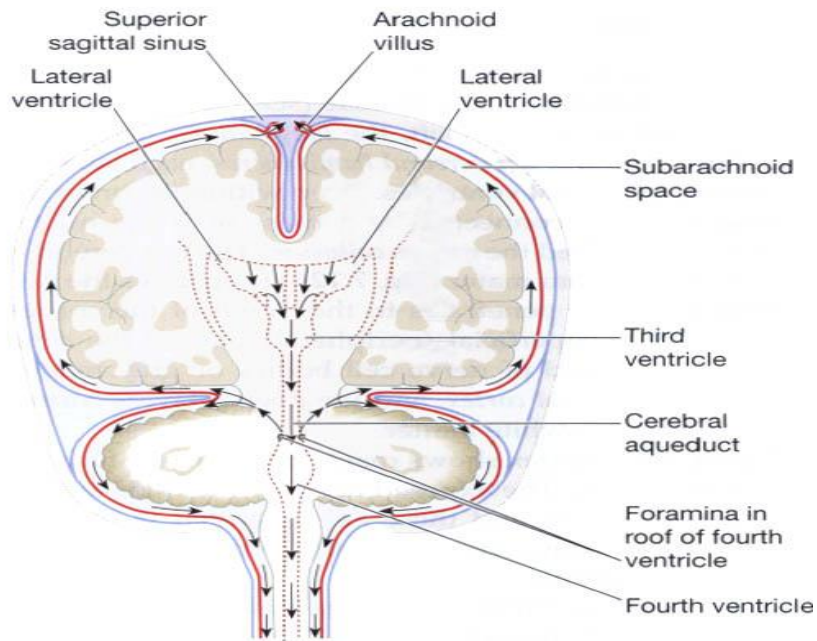
Cerebrospinal fluid (CSF)

- Colourless alkaline
- Composition-
 1. water
 2. mineral eg K, Cl, Na
 3. nutrients eg glucose, some proteins
 4. some leukocytes and globulin
 5. some creatinine and urea
- Specific gravity= 1.005
- Volume = 120mls (remaining constant)
- CSF is secreted continuously at a rate of about 0.5 ml per minute, i.e. 720 ml per day.

Ventricles of the brain and the cerebrospinal fluid

Within the brain, there are four irregular-shaped cavities, or *ventricles*, containing *cerebrospinal fluid*.

- right and left lateral ventricles.
- third ventricle
- fourth ventricle



The lateral ventricles

These cavities lie within the cerebral hemispheres, one on each side of the median plane just below the corpus callosum. They are separated from each other by a thin membrane, the septum lucidum, and are lined with ciliated epithelium. They communicate with the third ventricle by interventricular foramina.

The third ventricle

The third ventricle is a cavity situated below the lateral ventricles between the two parts of the thalamus. It communicates with the fourth ventricle by a canal, the cerebral aqueduct or aqueduct of the midbrain.

The fourth ventricle

The fourth ventricle is a diamond-shaped cavity situated below and behind the third ventricle, between the cerebellum and pons. It is continuous below the central canal of the spinal cord and communicates with the subarachnoid space by foramina on its roof. Cerebrospinal fluid enters the

subarachnoid space through these openings and the open distal end of the central canal of the spinal cord.

Functions of the CSF

- ✦ Support and protect the brain and the spinal cord.
- ✦ Acts as cushion and shock absorber b/n brains& cranial bones and b/n spinal cord & vertebra bones
- ✦ Is the medium of interchange of nutrients and waste products b/n nerve cells of the brain, spinal cord, and blood.
- ✦ It maintains uniform pressure around the brain and spinal cord.

Grey matter

The arrangement of grey matter in the spinal cord resembles the shape of the letter H, having two posterior, two anterior and two lateral columns.

Posterior columns of grey matter

These are composed of cell bodies which are stimulated by sensory impulses from the periphery of the body, it contributes to the formation of the white matter of the cord and transmit the sensory impulses upwards to the brain.

Anterior columns of grey matter

These are composed of the cell bodies of the lower motor neurones which are stimulated by the axons of the upper motor neurones or by the cell bodies of connector neurons linking the anterior and posterior columns to form reflex arcs.

The posterior root (spinal) ganglia are composed of cell bodies which lie just outside the spinal cord on the pathway of the sensory nerves.

White matter

The white matter of the spinal cord is arranged in three columns or tracts: anterior, posterior, and lateral. These tracts are formed by sensory nerve fibres ascending to the brain, motor nerve fibres descending from the brain and fibres of connector neurons.

Peripheral Nervous System

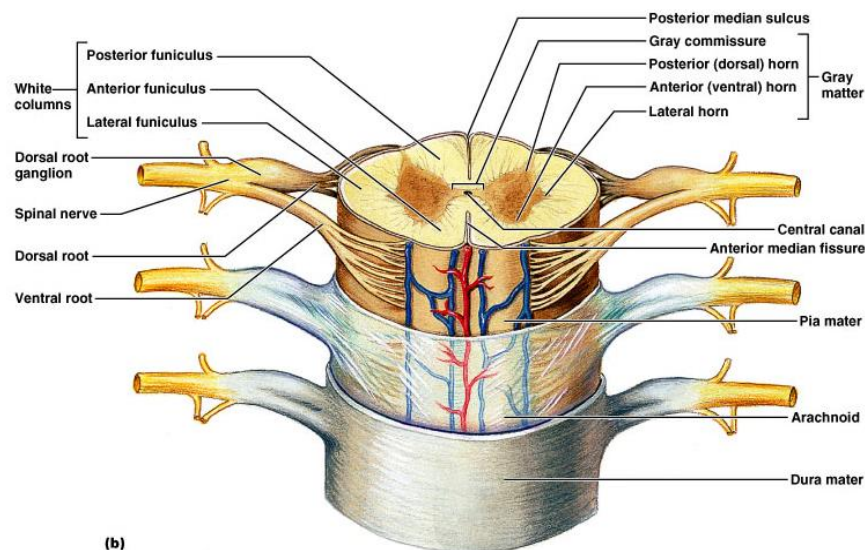
- Connects body to brain & spinal cord.
- 12 pairs of nerves from your brain (cranial nerves)
- 31 pairs from your spinal cord (spinal nerves)
 - Bundles of sensory and motor neurons held together by connective tissue.

- Most of the nerves of the peripheral nervous system are composed of sensory nerve fibres conveying afferent impulses from sensory end organs to the brain, and motor nerve fibres conveying efferent impulses from the brain through the spinal cord to the effector organs, e.g. skeletal muscles, smooth muscle, and glands.
- Each nerve consists of numerous nerve fibres collected into bundles. Each bundle has several coverings of protective connective tissue.
- Endoneurium is a delicate tissue, surrounding each fibre, which is continuous with the septa that pass inwards from the perineurium.
- Perineurium is a smooth connective tissue, surrounding each bundle of fibres.
- Epineurium is the fibrous tissue which surrounds and encloses several bundles of nerve fibres. Most large nerves are covered by epineurium.

Spinal nerves

- Part of the peripheral nervous system
- 31 pairs attach through dorsal and ventral nerve roots.
- Lie in intervertebral foramina.
 - 8 cervical
 - 12 thoracic
 - 5 lumbar
 - 5 sacral
 - 1 coccygeal.

Although there are only seven cervical vertebrae, there are eight nerves because the first pair leave the vertebral canal between the occipital bone and the atlas, and the eighth pair leaves below the last cervical vertebra.

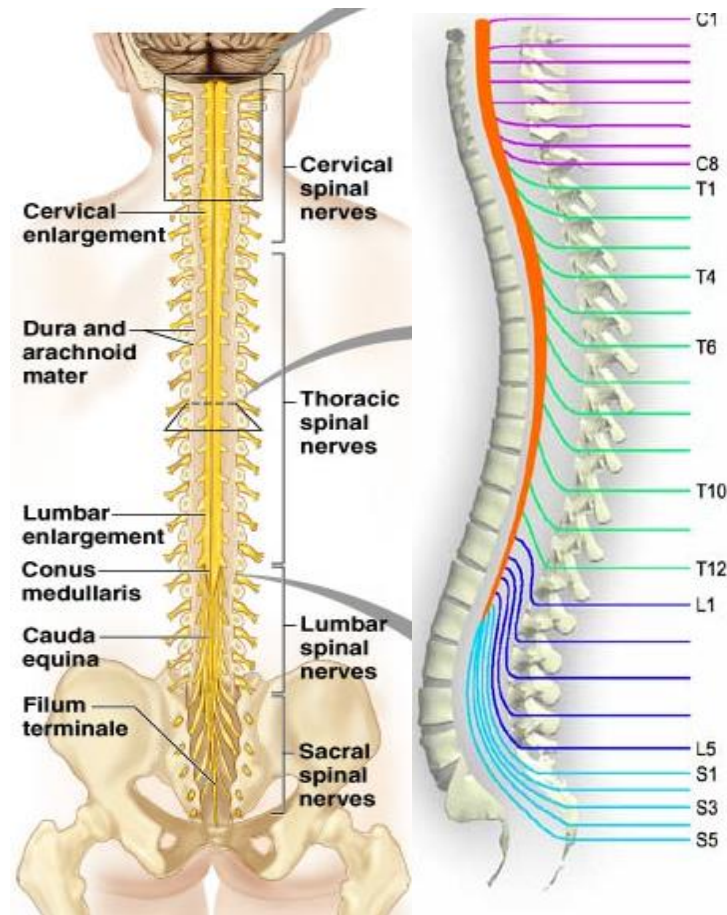


Spinal cord segments are superior to where their corresponding spinal nerves emerge through intervertebral foramina.

Spinal nerves are named according to the spinal cord segment from which they originate.

- 8 cervical
- 12 thoracic
- 5 lumbar
- 5 sacral
- 1 coccygeal

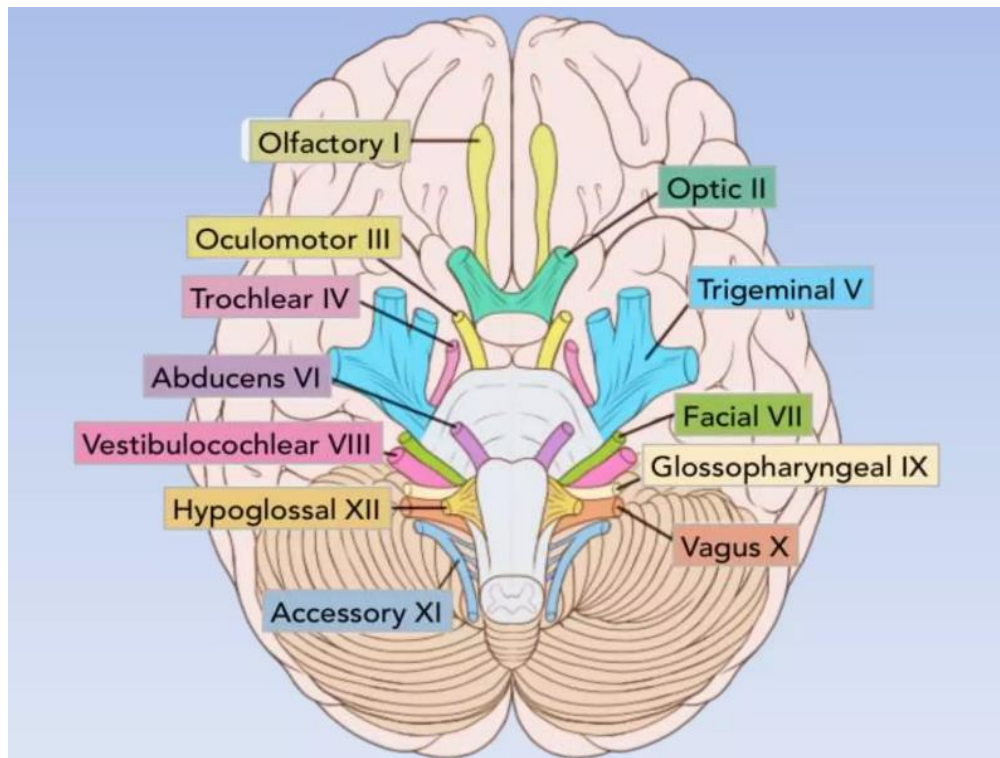
Cauda equina (“horse’s tail”): collection of nerve roots at the inferior end of the vertebral canal



Cranial nerves

- 12 pairs, Roman numerals I-XII
- Serve mainly head and neck.
 - Vagus – into thoracic and abdominal cavities
- All but the first 2 arise from the brain stem and pass through foramina in the base of the skull.
- Most are mixed (motor and sensory)
 - 3 are purely sensory:

- ✧ Optic
- ✧ Olfactory
- ✧ Vestibulocochlear



Number	Name	Modality	Function
1 (CN I)	Olfactory	Special Sensory	Smell
2 (CN II)	Optic	Special Sensory	Vision
3 (CN III)	Oculomotor	Somatic Motor Visceral Motor	Levator palpebrae, superioris, superior, medial & inferior recti muscles Parasympathetic to ciliary & pupillary constrictor muscles
4 (CN IV)	Trochlear	Somatic Motor	Superior oblique muscle
5 (CN V)	Trigeminal: – Ophthalmic – Maxillary – Mandibular	Branchial Motor General Sensory	Muscles of mastication Sensory for head/neck, sinuses, meninges, & external surface of the tympanic membrane
6 (CN VI)	Abducens	Somatic Motor	Lateral rectus muscle
7 (CN VII)	Facial	Branchial Motor Visceral Motor General Sensory Special Sensory	Muscles of facial expression Parasympathetic to all glands of head except the parotid Sensory for ear and tympanic membrane Taste the anterior two-thirds of the tongue
8 (CN VIII)	Vestibulocochlear	Special Sensory	Hearing and Balance
9 (CN IX)	Glossopharyngeal	Branchial Motor	Stylopharyngeus muscle

		Visceral Motor Visceral Sensory General Sensory Special Sensory	Parotid Gland Carotid Body The sensation of the posterior one-third tongue & internal surface of the tympanic membrane Taste posterior one-third tongue
10 (CN X)	Vagus	Branchial Motor Visceral Motor Visceral Sensory Special Sensory	Muscles pharynx & larynx Parasympathetic to neck, thorax, & abdomen Sensory from pharynx, larynx & viscera Sensory from the external ear
11 (CN XI)	Spinal accessory	Branchial Motor	Trapezius & sternocleidomastoid muscles
12 (CN XII)	Hypoglossal	Somatic Motor	Tongue muscles except palatoglossal

Functional Classification (Activities)

Afferent and Efferent Division

Afferent and efferent neurons refer to different types of neurons that make up the sensory and motor divisions of the peripheral nervous system, respectively. Neurons are electrically excitable cells that serve as the structural and functional unit of the nervous system.

The PNS has 2 subdivisions.

1. Sensory Division (Afferent) – conducts action potentials from PNS toward the CNS (by way of the sensory neurons) for evaluation.

Sensory afferent (from an organ to the CNS) fibers conveying impulses from the skin, skeletal muscles and joints are called **somatic afferent**, whereas those transmitting impulses from the visceral organs (thoracic, stomach, lungs, and urinary bladder) are called **visceral afferent** to CNS.

2. Motor Division (Efferent) – conducts action potentials from CNS toward the PNS (by way of the motor neurons) creating a response from an effector organ.

The motor (efferent –from CNS to organ) division transmits impulses from the CNS to effectors organs, the muscles, and the glands. The impulses activate muscles to contract and glands to secrete; that is, they effect (bring about), a motor response. The motor division in turn, has two subdivisions.

- a) Somatic Motor System – controls skeletal muscle only
Somatic nervous system. This allows us to consciously, or voluntarily, control our skeletal muscles. Relay information between skin, skeletal muscles, and central nervous system.
- b) Autonomic System – controls/effects smooth muscle, cardiac muscle, and glands
Autonomic nervous system (involuntary nervous system). Relay information from CNS to organs. The autonomic nervous system regulates events that are automatic, or involuntary. 2 branches of the autonomic system are:
 - Sympathetic – accelerator “fight or flight”
Sympathetic Nervous System: controls in times of stress, such as the flight or fight response.
 - Parasympathetic – brake “resting and digesting.”
Parasympathetic Nervous System: controls body in times of rest

REPRODUCTIVE SYSTEM

At the end of this lecture, students will know about the following:

1. Anatomy and physiology of the female and male reproductive system
2. The organs of each system
3. The functions of each organ in the system

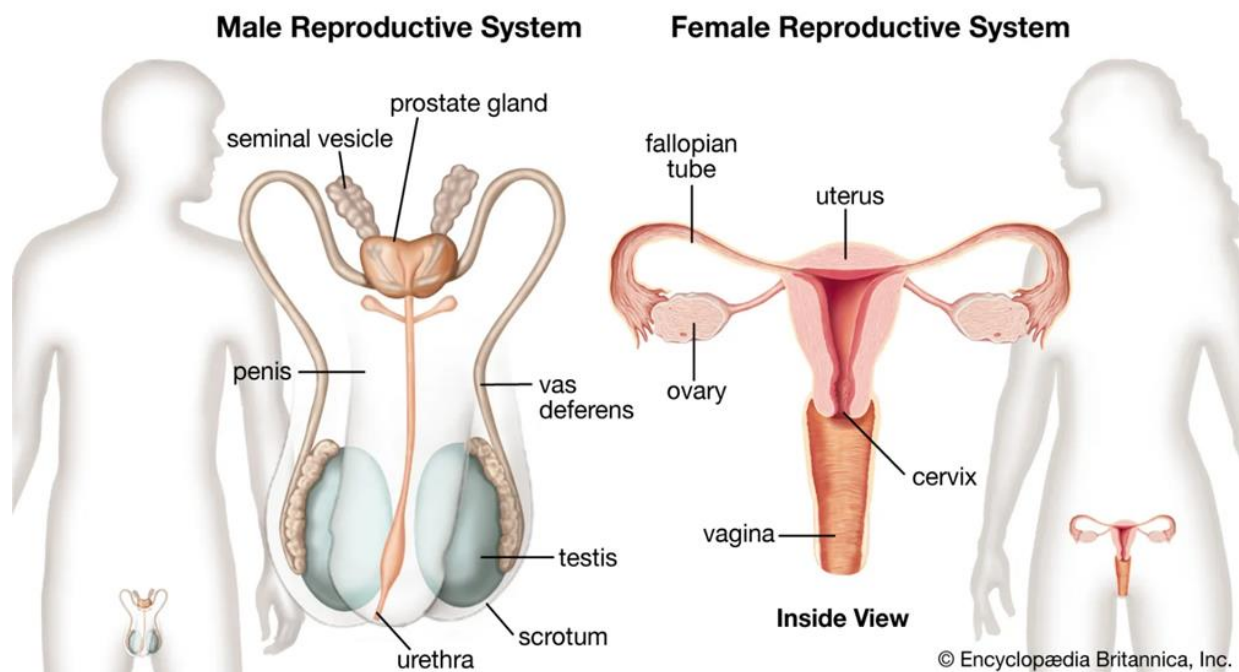
Content:

- ✚ Male reproductive system
- ✚ Endocrine system and male reproduction
- ✚ Spermatogenesis
- ✚ Female Reproductive System
- ✚ The Normal Menstrual Cycle
- ✚ Breasts

Reproduction can be defined as the biological process of producing a new individual or an offspring identical to the parents. This process ensures the increase in the number of individuals of a species when conditions are favourable. It is one of the fundamental characteristics of living things and an essential life process.

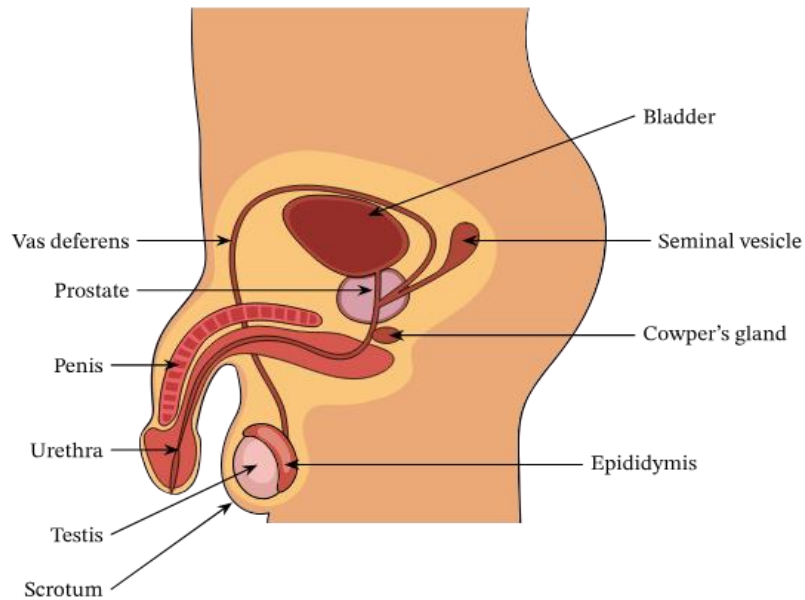
Human Reproductive System

The reproduction in human beings involves the fusion of male and female gametes produced in their reproductive system. The male reproductive system is different from the female reproductive system, both in structure and function. The male and female reproductive system functioning together produce a new life.



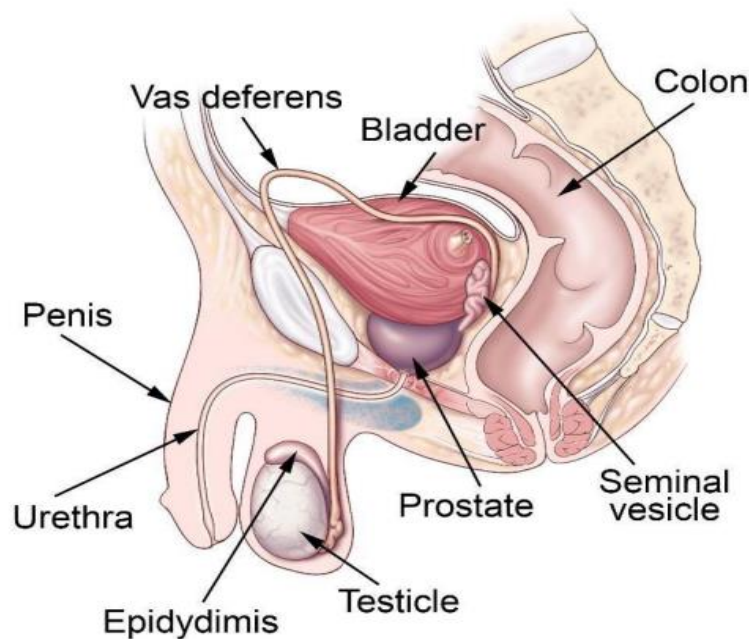
Male Reproductive System

The male gametes, i.e., sperms are produced within the male reproductive system. Sperms are small unicellular structures with a head, middle piece, and a tail.



Organs of the male reproductive system are divided into:

External and Internal Male Genitalia



External male Genitalia

Penis:

- Penis is a cylindrical tube which serves as both reproductive organ and an excretory organ. It delivers sperm into the vagina during sexual intercourse.
- It has three parts: **the root**, which attaches to the wall of the abdomen; **the body, or shaft; and the glans**, which is the cone-shaped end of the penis. The glans, which also is called the head of the penis, is covered with a loose layer of skin called foreskin. (This skin is sometimes removed in a procedure called circumcision.)
- The opening of the urethra, the tube that transports semen and urine, is at the tip of the glans penis.
- The penis also contains a number of sensitive nerve endings. The body of the penis is cylindrical in shape and consists of three internal chambers. These chambers are made up of special, sponge-like erectile tissue.
- This tissue contains thousands of large spaces that fill with blood when the man is sexually aroused.
- As the penis fills with blood, it becomes rigid and erect, which allows for penetration during sexual intercourse. The skin of the penis is loose and elastic to allow for changes in penis size during an erection.

Scrotum:

- It is a sac-like organ that hangs below the penis and behind it. It is the house of the testicles, or testes, and maintains a temperature that is required for the production of sperm by it.
- The scrotum has a protective function and acts as a climate control system for the testes.
- For normal sperm development, the testes must be at a temperature slightly cooler than the body temperature.
- Special muscles in the wall of the scrotum allow it to contract (tighten) and relax, moving the testicles closer to the body for warmth and protection or farther away from the body to cool the temperature.

Semen:

- Semen is a fluid which contains sperm and is expelled (ejaculated) through the end of the penis when the man reaches sexual climax (orgasm). When the penis is erect, the flow of urine is blocked from the urethra, allowing only semen to be ejaculated at orgasm.

- Semen nourishes the sperm, increases the volume and helps in lubrication.

Testicles (testes):

- A pair of oval-shaped organs masked in a pouch called the scrotum. They are responsible for the production of sperm and the male hormone testosterone.
- The testes are about the size of 5cm long and 2.5cm wide and lie in the scrotum, secured at either end by a structure called the spermatic cord.
- The testes are responsible for making testosterone, the primary male sex hormone, and for producing sperm.
- Within the testes are coiled masses of tubes called seminiferous tubules. These tubules are responsible for producing the sperm cells through a process called spermatogenesis.

The internal male reproductive organs (accessory organs)

Epididymis:

- The epididymis is a long, coiled tube that rests on the backside of each testicle. It functions in the carrying and storage of the sperm cells that are produced in the testes.
- It also is the job of the epididymis to bring the sperm to maturity, since the sperm that emerge from the testes are immature and incapable of fertilization.
- During sexual arousal, contractions force the sperm into the vas deferens.

Vas deferens:

- The sperms produced in the testes are stored in a tube called the epididymis. Here the sperms matured and pass to urethra through the muscular tube called the vas deferens.
- The vas deferens is a tube 45cm long, that travels from the epididymis into the pelvic cavity, to just behind the bladder.
- The vas deferens transports mature sperm to the urethra in preparation for ejaculation.

Ejaculatory ducts:

- These are formed by the fusion of the vas deferens and the seminal vesicles. The ejaculatory ducts empty into the urethra.

Urethra:

- The urethra is the tube that carries urine from the bladder to the outside of the body.
- In males, it has the additional function of expelling (ejaculating) semen when the man reaches orgasm.

- When the penis is erect during sex, the flow of urine is blocked from the urethra, allowing only semen to be ejaculated at orgasm.

Seminal vesicles:

- The seminal vesicles are sac-like pouches that attach to the vas deferens near the base of the bladder.
- The seminal vesicles produce a sugar-rich fluid (fructose) that provides sperm with a source of energy and helps with the sperm's motility (ability to move).
- The fluid of the seminal vesicles makes up most of the volume of a man's ejaculatory fluid, or ejaculate.

Prostate gland:

- The prostate gland is a walnut-sized structure that is located below the urinary bladder in front of the rectum.
- The prostate gland contributes additional fluid to the ejaculate. Prostate fluids also help to nourish the sperm.
- The urethra, which carries the ejaculate to be expelled during orgasm, runs through the centre of the prostate gland.

Bulbourethral glands

- The bulbourethral glands, or Cowper's glands, are pea-sized structures located on the sides of the urethra just below the prostate gland. These glands produce a clear, slippery fluid that empties directly into the urethra.
- This fluid serves to lubricate the urethra and to neutralize any acidity that may be present due to residual drops of urine in the urethra.

Endocrine system and male reproduction

The entire male reproductive system is dependent on hormones, which are chemicals that stimulate or regulate the activity of cells or organs. The primary hormones involved in the functioning of the male reproductive system are follicle stimulating hormone (FSH), luteinizing hormone (LH) and testosterone.

FSH and LH are produced by the pituitary gland located at the base of the brain. FSH is necessary for sperm production (spermatogenesis), and LH stimulates the production of testosterone, which is necessary to continue the process of spermatogenesis. Testosterone also is important in the

development of male characteristics, including muscle mass and strength, fat distribution, bone mass and sex drive.

Spermatogenesis

It is the formation of sperm process begins at puberty and continues during the male's life. A sperm's fertile life is 5 days after ejaculation. Sperm are much smaller than ova. Sperm cells resemble tadpoles in shape with oval heads and long tails. During each ejaculation approximately 300 million sperm are deposited in to the vagina. only few reach the ova and just one sperm penetrates and fertilizes the ovum.

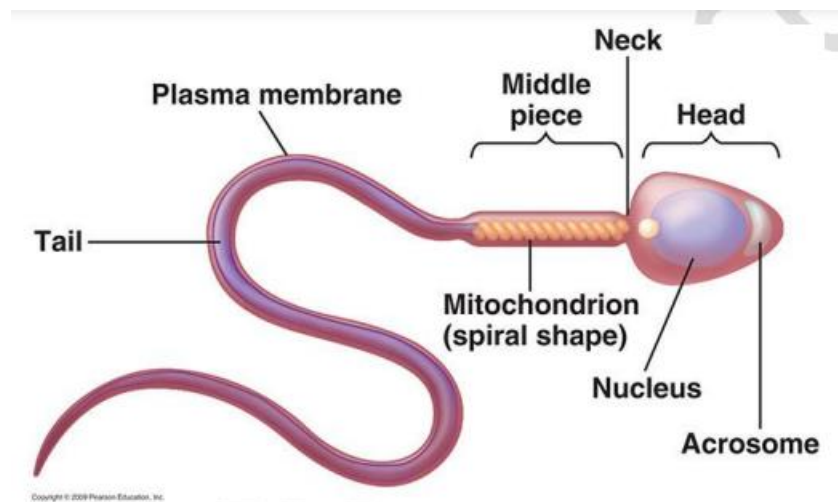


Figure (10) the sperm

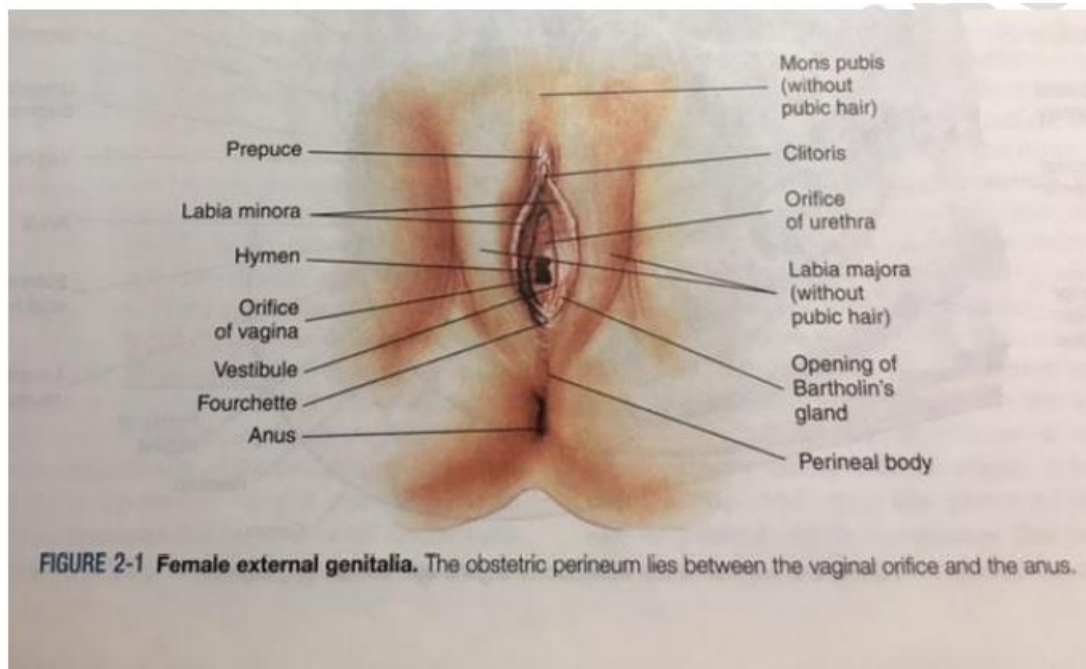
Female Reproductive System

The female reproductive system is active before, during and after fertilization as well.

It consists of the following parts:

- **A pair of ovaries:** Ovaries produce and store ovum in them. They also produce a female hormone called estrogen.
- **Fallopian tubes (Oviducts):** They are the site of fertilization. They connect the ovaries with the uterus.
- **Uterus:** The uterus is the site of development for the embryo.
- **Vagina:** It is the part which connects the cervix to the external female body parts. It is the route for the penis during coitus as well as a fetus during delivery.

External female Genitalia:



Vulva

- The female external reproductive organs consist of the mons pubic, which is covered with pubic hair; two folds of tissue, called the labia majora and labia minora, which surround a space called vestibule.

Mons Pubis

- The mons pubis is formed at the upper margin of the symphysis pubis and is shaped like an inverted triangle.
- It is located over the two pubic bones of the pelvic.
- This structure is composed of fatty tissue lying beneath the skin and from puberty on, is covered with varying amount of hair.
- The mons pubis surrounds delicate tissue and protects it from injury.

Labia Majora and Labia Minora

The labia majora:

- are two folds of fatty tissue that form the lateral boundaries of the vulva.
- They are covered with coarse skin and pubic hair on the outer aspect and are smooth and moist on the inner aspect, where the openings of numerous small glands are found.

The labia minora:

- are soft folds of skin that are rich in sebaceous glands.
- The labia minora are moist and are composed of erectile tissue containing loose connective tissue, blood vessels, and involuntary muscles.
- The functions of the labia minora are to lubricant and waterproof the vulvar skin and to provide bactericidal secretion that help prevent infections.

Clitoris

- The clitoris is a small, sensitive structure that, like the penis, is composed of erectile tissue, nerves, and blood vessels; it is covered at the tip with very sensitive tissue.
- It exists primarily for female sexual enjoyment.

Vaginal Vestibule

- The vaginal vestibule is a boat-shaped depression enclosed by the labia minora and is visible when the labia minora are separated.
- The vestibule contains the vaginal opening (introitus), which is located between the external and internal genitalia.
- The vestibule contains the openings of five structures that drain into it the ureteral meatus, skene's ducts, and the ducts from Bartholin's glands that are located on each side of the vagina.
- The vestibule ends with the formation of the fourchette.

Urethra

- The opening to the urethra is just below the clitoris.
- Although it is not related to sex or reproduction, it is included in the vulva.
- The urethra is actually used for the passage of urine.
- The urethra is connected to the bladder.
- In females the urethra is 1.5 inches long, compared to males whose urethra is 8 inches long.

Hymen

- The hymen is a thin, elastic, mucous membrane that partially covers the vagina in young females.
- Does not seem to have a specific physiological function or purpose.
- Many shapes are possible.

- Normal variations of the hymen range from thin and stretchy to thick and somewhat rigid; or it may also be completely absent

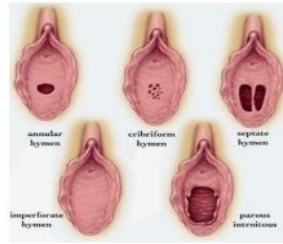


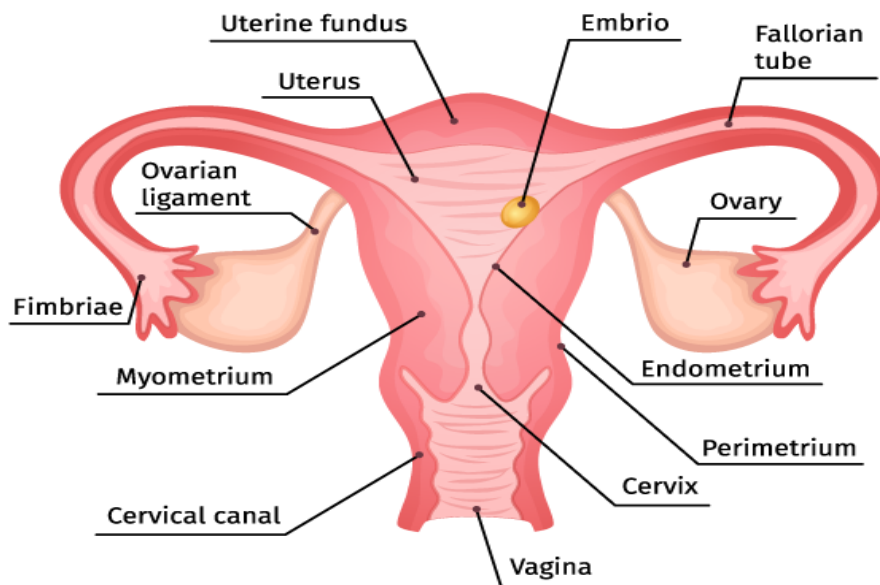
Figure (1) Types of hymen

Perineum

- The perineum is the short stretch of skin starting at the bottom of the vulva and extending to the anus.
- It is a diamond-shaped area between the symphysis pubis and the coccyx.
- This area forms the floor of the pelvis and contains the external sex organs and the anal opening.
- The perineum in some women may tear during the birth of an infant and this is apparently natural.
- Some physicians may cut the perineum preemptively on the grounds that the "tearing" may be more harmful than a precise cut by a scalpel.
- If a physician decides the cut is necessary, they will perform it. The cut is called an episiotomy.

Internal Female Genitalia

FEMALE REPRODUCTIVE SYSTEM



Vagina

The vagina is a curved tube leading from the uterus to the external opening at the vestibule. It lies between the urinary bladder and the rectum. Because it meets at a right angle with the cervix, the interior wall is about 2.5cm(1 inch) shorter than posterior wall, which varies from 7-10cm (approximately 2.8 to 4 inches). It consists of muscle and connective tissue and is lined with epithelial tissue, which contains folds called rugae. These folds allow the vagina to stretch considerably during childbirth. The epithelial cells lining the vagina show cyclic changes related to estrogens, progestins, and androgens. Doderlein's bacilli, which are normally present in the vagina, act on glycogen from the epithelial cells to produce lactic acid. This maintains the acidity of the vagina and is the reason that the vagina is resistant to most infections. A change in the PH of the vagina, which can be caused by frequent douching, antimicrobial therapy, or deodorant tampons, can increase the vagina's susceptibility to invading pathogens.

Purposes of the Vagina

- Receives a male's erect penis and semen during sexual intercourse.
- Pathway through a woman's body for the baby to take during childbirth.
- Provides the route for the menstrual blood (menses) from the uterus, to leave the body.
- May hold forms of birth control, such as a diaphragm or female condom.

Cervix

The cervix consists of a cervical canal with an internal opening near the uterine corpus called the internal os and an opening into the vagina called external os the mucosal lining of the cervix has four functions:

- Providing lubrication for the vagina
- Acting as bacteriostatic agent
- Providing an alkaline environment to shelter the sperm from the acidic vagina
- Producing a mucous plug in the cervical canal during pregnancy

Uterus

- The uterus (womb) is a hollow, pear-shaped, muscular organ. It is approximately 2.5 cm (1 inch) thick, 5cm (2 inches) wide, and 7.5cm (3 inches) long.
- During pregnancy, the uterus can stretch and enlarge considerably. the weight of the non-pregnant uterus is approximately 75g; it increases to approximately 907g during pregnancy, the uterus increases in vascularity, which allows sufficient blood supply for its growth, and can stretch and enlarge to a considerable size.
- After pregnancy, it returns almost entirely to its former weight, size, and shape.
- The uterus lies between the bladder and the rectum. It is supported by two important pairs of ligaments, the round and broad ligaments.
- The uterus is divided into three parts: fundus (upper portion), the corpus (body), the cervix
- The uterus has three layers (perimetrium, myometrium, endometrium).

The following three functions of the uterus:

1. Menstruation: the uterus sloughs off of the endometrium or lining of the uterus.
2. Pregnancy: the uterus supports the fetus and allows fetus to grow.
3. Labor and birth: the uterine muscle contract and the cervix dilates during labor to expel the fetus

Fallopian Tube

- The fallopian tubes extend laterally from the uterus, one to each ovary.
- They are small, narrow and approximately 10cm (4 inches) long.
- The tubes carry the ovum from the ovary to the uterus by the contraction of the cilia: hair-like projections found in the lining of the tubes.
- Extending from the ends of the fallopian tubes are small, fingerlike projections called fimbriae.
- Their movement sweeps the ovum into the tube, after which the ovum travels to the uterus. It takes approximately 5 days for the ovum to travel the 10 cm from the ovary to the uterus.
- Fertilization of the ovum with sperm normally takes place in the outer third of the fallopian tube.

The four functions of the fallopian tube:

1. A passageway in which sperm meet the ovum
2. A site of fertilization
3. A safe nourishing environment for the ovum or zygote (fertilized ovum)
4. A means of transport ting the ovum or zygotes to the corpus of the uterus

Ovaries

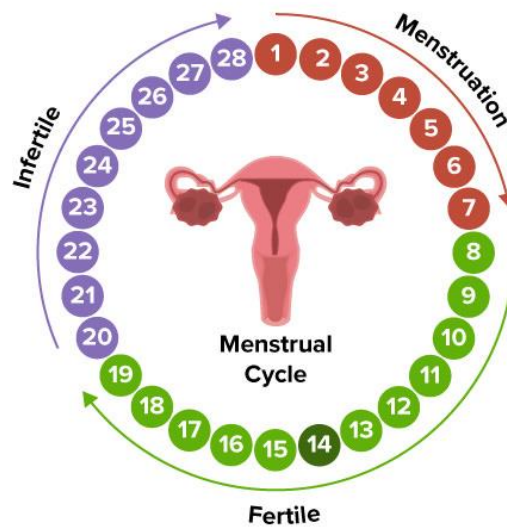
- The ovaries in the female and the testes in the male are similar in embryologic origin.
- The ovaries are two small, almond-shaped organs located on each gland.
- Approximately 2 million ova are present at birth. Many ova degenerate until puberty, when a few thousand remain.
- During the course of a woman's reproductive life, only about 400 ova mature enough to be fertilized.
- During each menstrual cycle one follicle matures into what is called a graafin follicle, which contains the ovum that is released each month during ovulation.
- Estrogen released by the ovary stimulates the development of secondary sexual characteristics such as the breasts.
- Progesterone is responsible for preparing and maintaining the lining of the uterus for implantation of the ovum.

The ovaries have two functions:

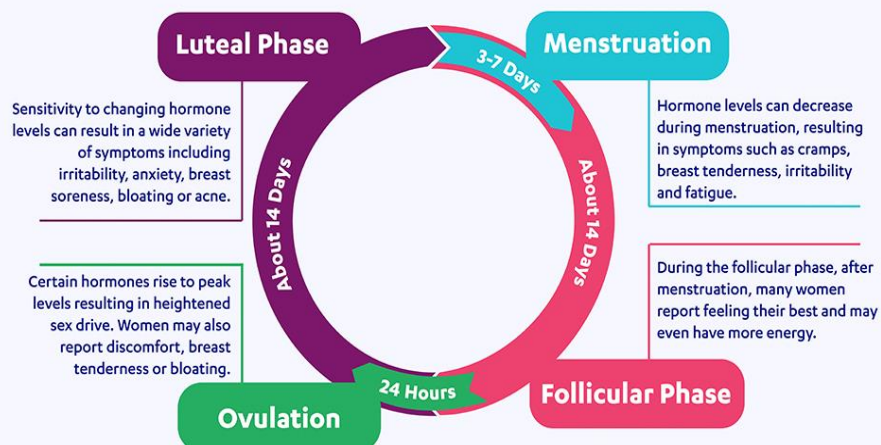
1. The development and maturation of and later expulsion of the ovum (ovulation)
2. The secretion of hormones (estrogen & progesterone)

The Normal Menstrual Cycle

The menstrual cycle is a predictable event that normally occurs monthly. The typical monthly menstrual cycle is influenced by ovarian follicle maturation, ovulation and corpus luteum formation and ends with menstrual bleeding. The changes that occur depend on the changes occurring simultaneously in the ovaries. The development of endometrium occurs at the precise time of the month that the release of a mature ovum occurs. Ovulation occurs 14 days before the beginning of the next menstrual cycle. And the ovum remains fertile for 24 hours. The sperm can survive up to 5 days. Fertilization most often occurs in the first few hours after ovulation.

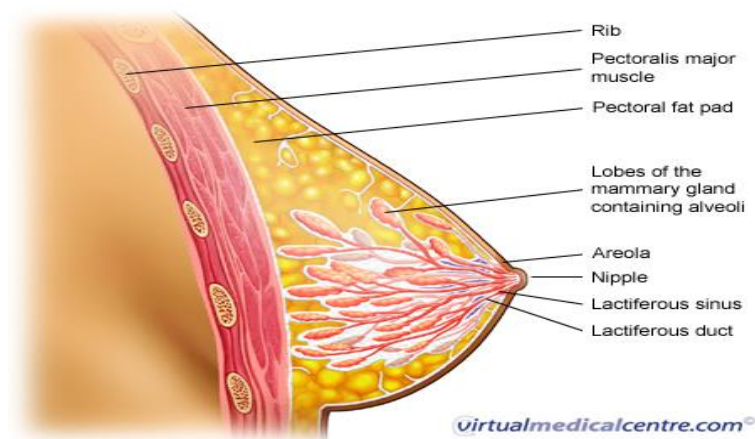


What To Know About Your Menstrual Cycle



Breasts

The breast is composed of glandular, ductal, connective, and adipose tissue. Embedded in the fibrous tissue are fat and lobules which make up the mammary glands, accessories to reproduction in women, but rudimentary and functionless in men. In men, little fat is present in the breast, and the glandular system normally does not develop. In women, the breasts are the most prominent superficial structure on the anterior thoracic wall, and the amount of fat in the glandular tissue determines the size of the breasts. A small part of the mammary gland often extends into the axilla, forming the axillary tail of Spence.



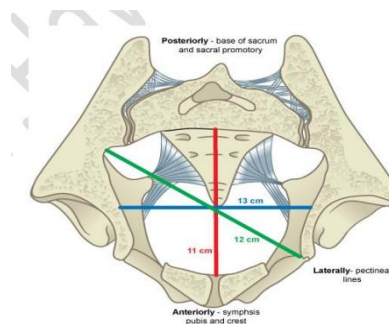
The mammary glands are modified sweat glands and are composed of 15-20 lobules, each drained by a lactiferous duct. Each lactiferous duct independently drains on the nipple and is preceded by a small, dilated portion known as the lactiferous sinus. It is in the sinus that milk collects during nursing and is "let down" by the suckling action of the infant the process is called lactation the nipple in the center of breasts is surrounding by pigmented areola, which darkens during pregnancy . Each breast consists of 10-20 lobes each lobe divided into 20- 40 lobules each lobule divided into 20-80 alveoli.

During pregnancy high levels of estrogen and progesterone produced by the placenta inhibit milk secretion after the expulsion of the placenta there is an abrupt change in estrogen and progesterone levels. This allows a hormone called prolactin to be released from the anterior pituitary gland when the infant sucks. Prolactin stimulates produce the milk the infant sucking and also stimulates releasing oxytocin hormone from the posterior pituitary gland that causes eject the milk from the

alveoli into the ductal system. The size of the breast depends on the amount of fatty tissue in the breast.

Pelvis

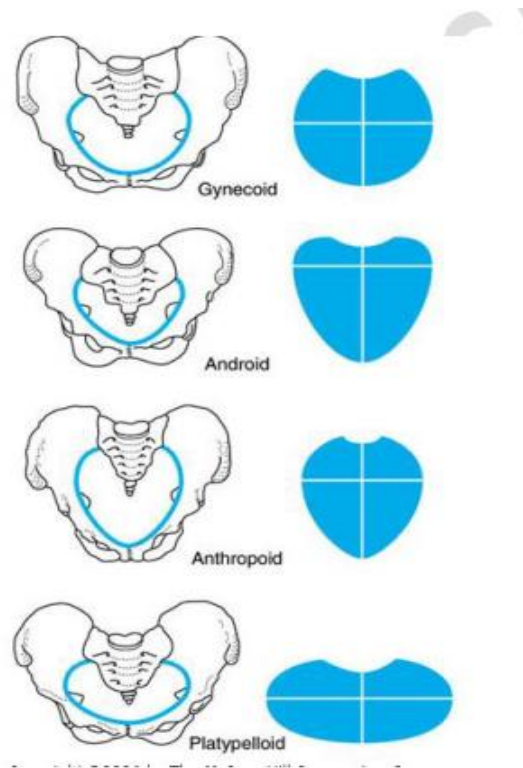
The composition of hip bone the hip bone is made up of three parts – the ilium, pubis and ischium. The superior part of the hip bone is formed by the ilium, the widest and largest of the three parts. The pubis is the most anterior portion of the hip bone. The posterior inferior part of the hip bone is formed by the ischium. The pelvic cavity is divided into sections, the false and true pelvic, the two ilia form the upper part of the pelvis known as the false pelvic. The ischial spines' sharp projections that form the posterior border of the ischium are important landmarks and represent the shortest distance of the pelvic cavity. The true pelvic (lower part) consists of the inlet, pelvic cavity and outlet and is most important during birth.



Types of the pelvic

The pelvic is divided into four types:

1. Gynaecoid (50%): normal female-type pelvic which is round
2. Anthropoid (20%): which has a long antero-posterior outlet
3. Android (20%): male-type pelvic which has heart heart-shaped outlet
4. Platypelloid (5%): which has a wide transvers outlet and is not favorable to a vaginal delivery.



Within the context of producing offspring, the reproductive system has four functions:

- To produce egg and sperm cells
- To transport and sustain these cells
- To nurture the developing offspring
- To produce hormones

THE RESPIRATORY SYSTEM

At the end of this lecture, students will know about the following:

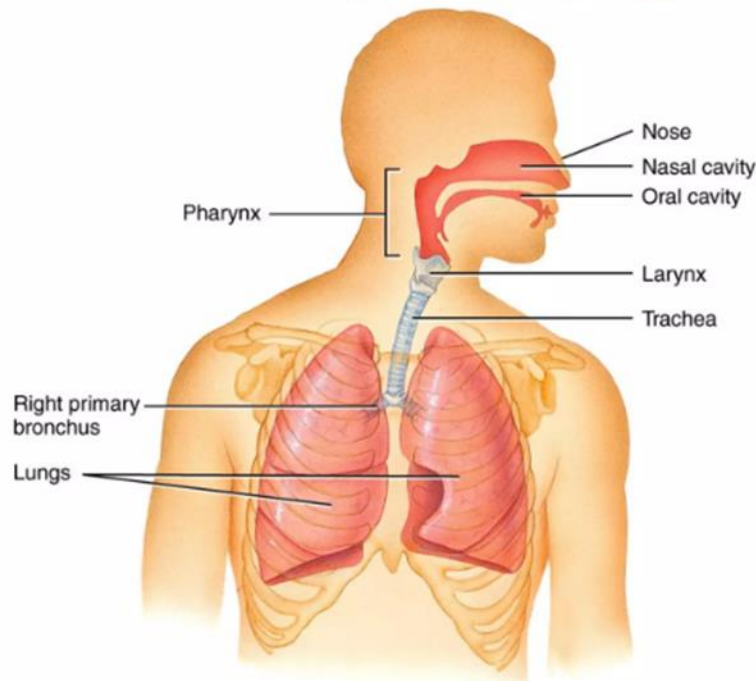
1. Anatomy and physiology of the respiratory system
2. The organs of the system
3. The functions of each organ in the system

The Respiratory System is the network of organs and tissues that help you breathe. It includes your airways, lungs, and blood vessels. The muscles that power your lungs are also part of the respiratory system. These parts work together to move oxygen throughout the body and clean out waste gases like carbon dioxide.

Respiration is the act of breathing:

- **Inhaling.** The act of breathing in oxygen.
- **Exhaling.** The act of breathing out carbon dioxide.

Organs of the Respiratory System



Anterior view showing organs of respiration

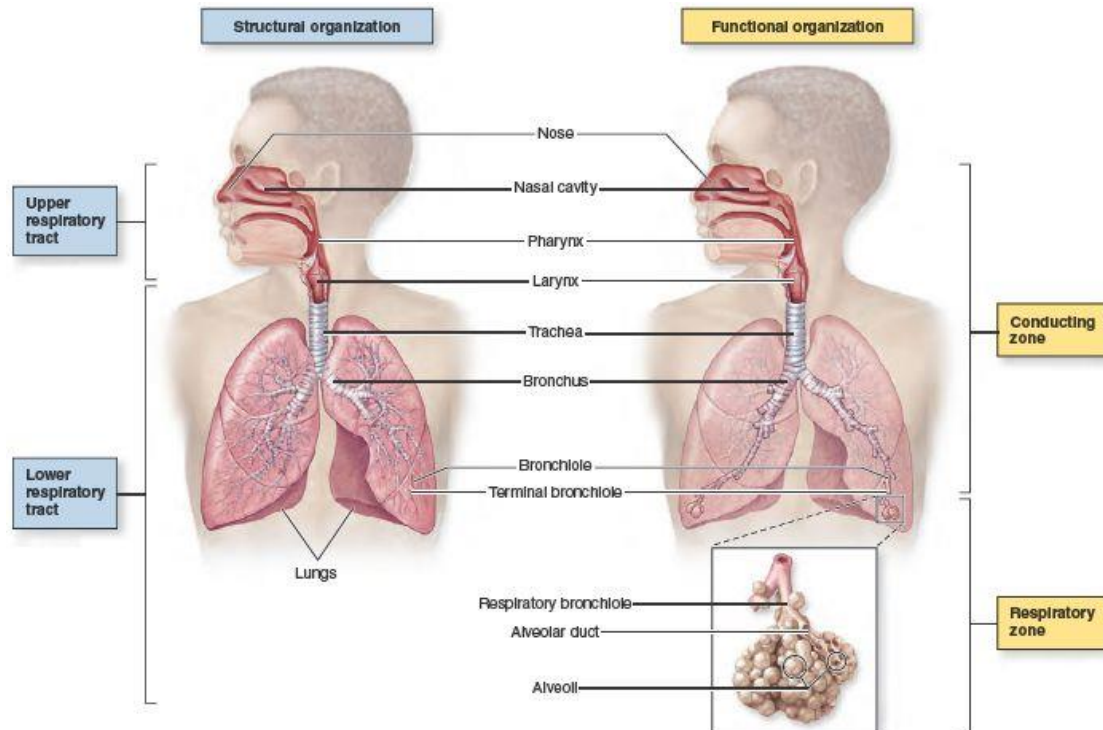
The 3 major steps in respiration

- Pulmonary ventilation** - (moving air in and out of the lungs)
External respiration - (gas exchange between alveoli and blood)
Internal respiration - (gas exchange between blood and cells)

Organs of the Respiratory System

Upper respiratory system - (nose, pharynx, Sinuses)

Lower respiratory system- (trachea, larynx, bronchi, bronchioles, and lungs)



The conducting zone consists of all airways that carry air to the lungs (Nose, pharynx, trachea, larynx, bronchi, bronchioles and terminal bronchioles)

Respiratory zone: Sites within the lungs where gas exchange occurs (Respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli)

Upper Respiratory System:

1. Nose

Your nose is part of your respiratory system. It allows air to enter your body, then filters debris and warms and moistens the air. Your nose gives you a sense of smell and helps shape your appearance.

Structure

- External nares → nasal cavity → internal nares

- Nasal septum divides nose into two sides
- Nasal conchae covered by mucous membrane

Function

Your nose is involved in several important bodily functions:

- Allows air to enter your body.
- Contributes to how you look and how you sound when you speak.
- Filters and cleans air to remove particles and allergens.
- Provides a sense of smell.
- Warms and moistens air so it can move comfortably into your respiratory system.

2. Pharynx (Throat)

The pharynx, commonly called the throat, is a passageway that extends from the base of the skull to the level of the sixth cervical vertebra. It serves both the respiratory and digestive systems by receiving air from the nasal cavity and air, food, and water from the oral cavity.

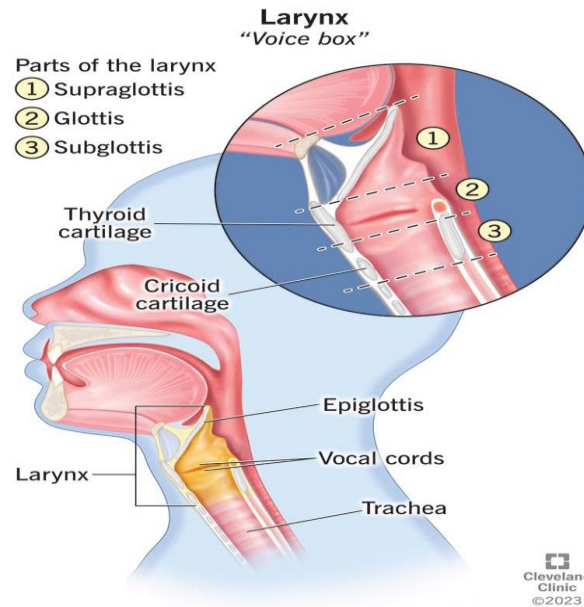
- It is a 12-14 cm long tube which lies behind the nose (nasopharynx), the mouth (oropharynx), and the larynx (laryngopharynx)
- The pharynx is lined with mucous membrane and has a smooth muscle layer which helps in swallowing

Functions

- Passage for air and food
- Warming and humidifying (like the nose)
- Taste, via nerve endings in the oropharynx
- Hearing, the auditory tube from the middle ear opens into the nasopharynx allowing air to enter the middle ear which is critical for hearing
- The tonsils give protection against infection
- Help speech by acting as a resonating chamber for the larynx

Upper Respiratory System

1. Larynx (voice box)



Your larynx is part of your respiratory system. It's a hollow tube that's about 4 to 5 centimeters (cm) in length and width. It lets air pass from your throat (pharynx) to your trachea on the way to your lungs. Your larynx is also the reason you're able to make sounds, so it's often called your voice box.

Your larynx is divided into three parts:

- The upper part (supraglottis).
- The middle part (glottis).
- The lower part (subglottis).

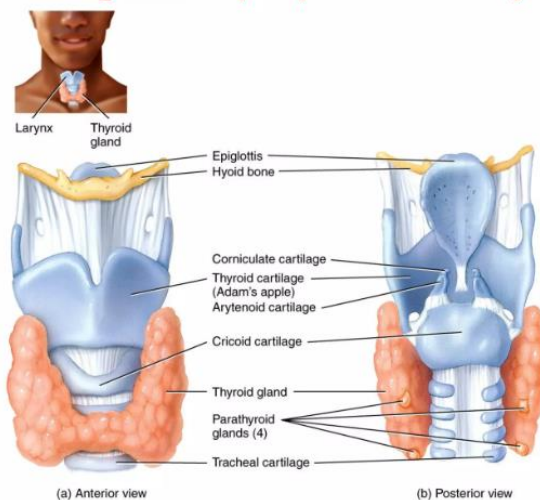
The voice box is made largely of cartilage:

- Thyroid cartilage; V-shaped
- (Adams's apple, projects more anteriorly in males after puberty in boys, hence the Adam's apple is more prominent in men)
- Epiglottis; leaf-shaped piece, cover airway
- The voice cords are found inside the larynx. Air passing up from the lungs vibrates the cords producing sound.
- The pitch varies depending on the length and tightness of the cords and the volume depends on the force with which the cords vibrate.

Function

- Warming, humidifying and filtering
- Forms a passageway for air between the pharynx and trachea
- Production of sound and speech. The sounds produced by the vocal cords are manipulated by the muscles of the tongue, cheeks and lips.
- Protects the lower respiratory tract, during swallowing the epiglottis closes off the respiratory tract allowing food to pass down the esophagus

2. The Trachea (Windpipe)



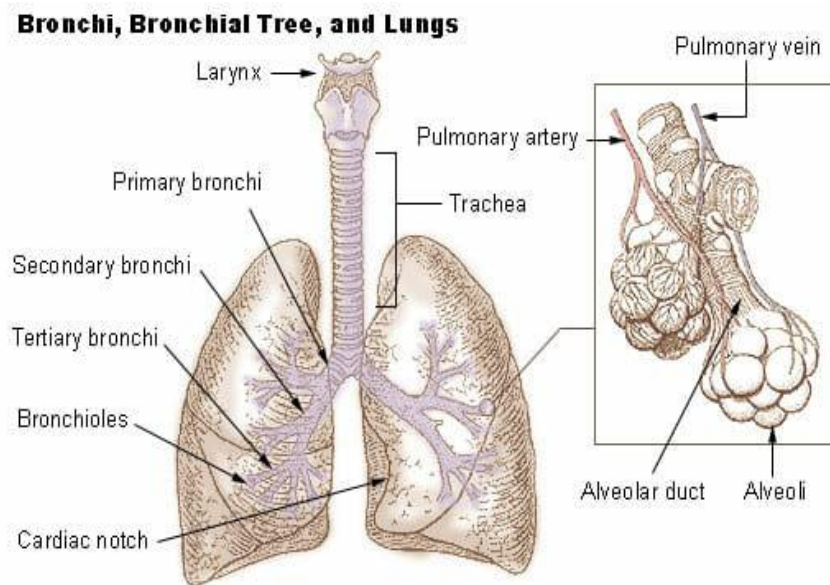
The trachea is a tube-like structure supported by C-shaped ring of cartilage stacked on top of each other in the neck and upper chest. It stretches from the larynx, or voice box, to the bronchi and transports air to and from the lungs when a person breathes. When a person inhales, air travels through the nose or mouth, down the trachea, and into the lungs.

At the level of the 5th thoracic vertebrae, the trachea divides into the left and right bronchi, the bronchus going to each lung.

Functions

- Warming, humidifying and filtering
- Allows the passage of air from the upper respiratory tract to the lungs
- Protection, the trachea is lined with cilia that move foreign particles up to the larynx where they are either swallowed or coughed up. The coughing reflex involves nerve endings in the larynx, trachea and bronchi.

3. The Bronchi (Large airways)



Your bronchi are the large tubes that connect to your trachea (windpipe) and direct the air you breathe to your right and left lungs. They are in your chest. Bronchi is the plural form of bronchus. The left bronchus carries air to your left lung.

Function

- Your bronchi carry air to and from your lungs.
- The bronchi also help moisturize the air you breathe and screen out foreign particles.
- Your airways are lined with cells that create mucus. The mucus keeps your airways moist. It also traps bacteria, viruses, fungi and other particles to protect your lungs and prevent infection.
- The bronchi are lined with cilia, tiny hairlike structures. The cilia help move mucus (phlegm) and particles out of your lungs. When you cough or swallow, the particles trapped in the mucus move out of your body or into your digestive tract, where your body can dispose of them.

3. Bronchioles (Small airways)

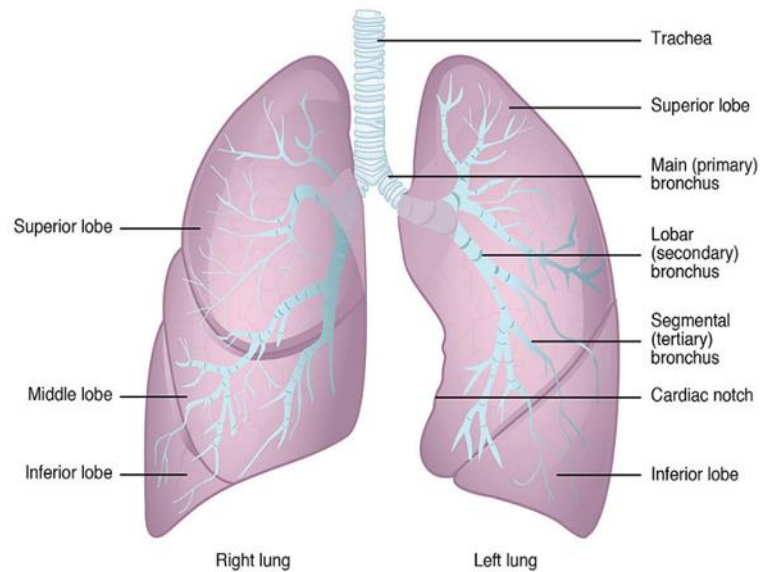
Small branches of the bronchial tubes that lead to the alveoli.

Bronchioles are air passages inside the lungs. They branch off like tree limbs from the bronchi. The two main air passages through which air flows from the trachea (windpipe) after being inhaled through the nose or mouth.

Function

The function of the bronchioles is to deliver air to a diffuse network of around 300 million alveoli in the lungs.

4. The Lungs



The lungs are a pair of cone-shaped organs made up of spongy, pinkish-grey tissue. They take up most of the space in the chest (thorax).

The lungs are the major organs of the respiratory system and are divided into sections or lobes. The right lung has three lobes and is slightly larger than the left lung, which has two lobes. The lungs are separated by the mediastinum. This area contains the heart, trachea, oesophagus, and many lymph nodes.

The lungs are covered by a protective membrane known as the pleura and are separated from the abdominal cavity by the muscular diaphragm

Blood Supply

The lungs are supplied with blood by both pulmonary and bronchial circulation. Pulmonary circulation carries deoxygenated blood from the heart to the lungs and transports back oxygenated blood to the heart that can be supplied to the body. The bronchial circulation,

on the other hand, transports oxygenated blood to the airway of the lungs with the help of bronchial arteries that come from the aorta.

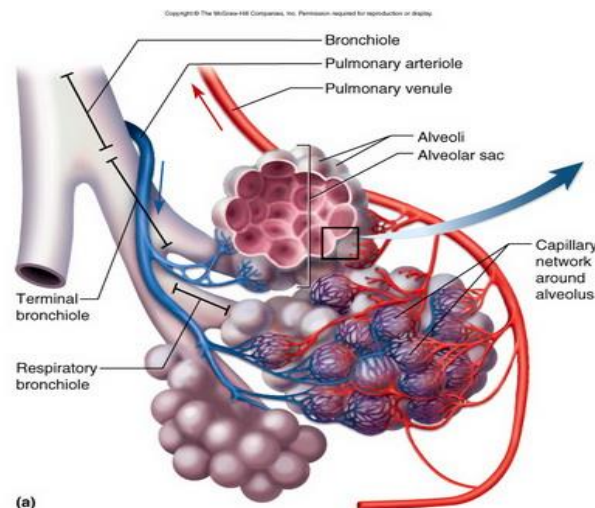
Functions of the lungs

- **Gaseous Exchange:** The primary function of the lungs is to facilitate gaseous exchange. Alveoli, the functional unit of the lungs, have thin membranes that allow easy exchange of gases. However, the lungs cannot expand or relax on their own during the breathing process. The diaphragm and intercostal muscles are the two main respiratory muscles that help in inhalation and exhalation.

These muscles contract, pull up the rib cage and increase the volume of the lungs for inhalation. During breathing out, the diaphragm and intercostal muscles relax, bringing the rib cage back to its original position.

- **Protection:** The respiratory tract is lined with respiratory epithelium that is interspersed with goblet cells. The goblet cells secrete mucus which acts as an important barrier against airborne infections. The mucus also contains many antimicrobial compounds such as antioxidants, antiproteases, and defensins.
- **Other Functions:** The lungs help in regulating blood pressure by becoming a part of the renin-angiotensin complex. It also maintains an acid-base homeostasis in the blood by expelling carbon dioxide during breathing.
- The fluid between the pleural membranes lubricates the pleural surface so they slide against each other smoothly when the lungs expand and contract.

5. Alveoli



Tiny air sacs in the lungs where the exchange of oxygen and carbon dioxide takes place.

Alveoli, also known as air space or air sacs, are millions of hollow cup-shaped cavities found in the lungs where the exchange of oxygen and carbon dioxide takes place. The alveoli make up the functional tissue of the lungs called lung parenchyma which occupies around 90% of the total lung space.

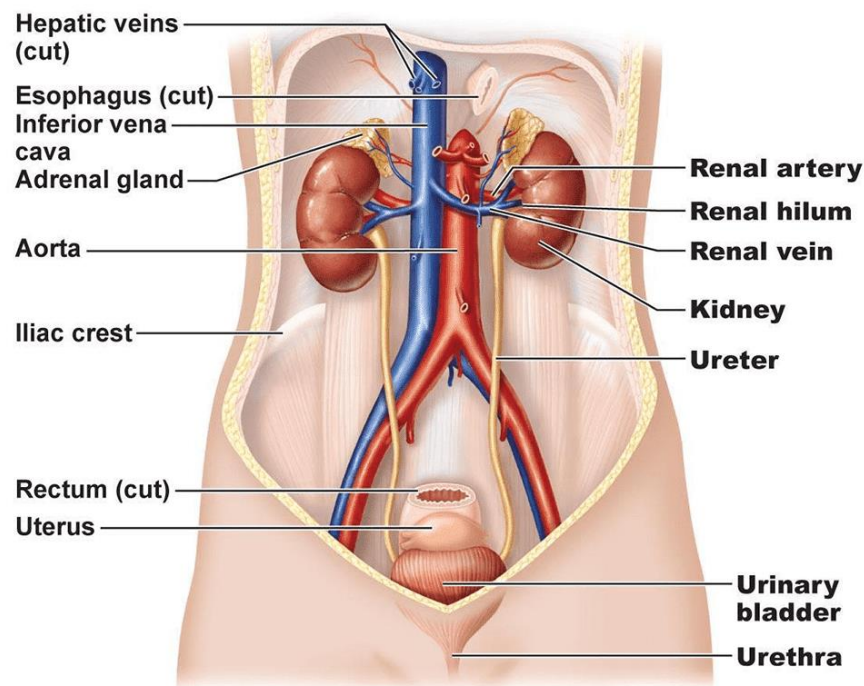
THE URINARY SYSTEM

Thousands of metabolic processes in myriad body cells produce hundreds of waste products. The urinary system removes them by filtering and cleansing the blood as it passes through the kidneys.

The urinary system consists of 4 major organs: the kidneys, ureters, urinary bladder, and the urethra. Together these organs act to filter blood, remove waste products, create urine, and transport urine out from the body. The urinary system is also called the excretory system, because held within the urine are the various excreted products, including by-products such as urea and uric acid, drugs, excess body fluid and excess salts (electrolytes).

- Paired kidneys
- A ureter for each kidney
- Urinary bladder
- Urethra

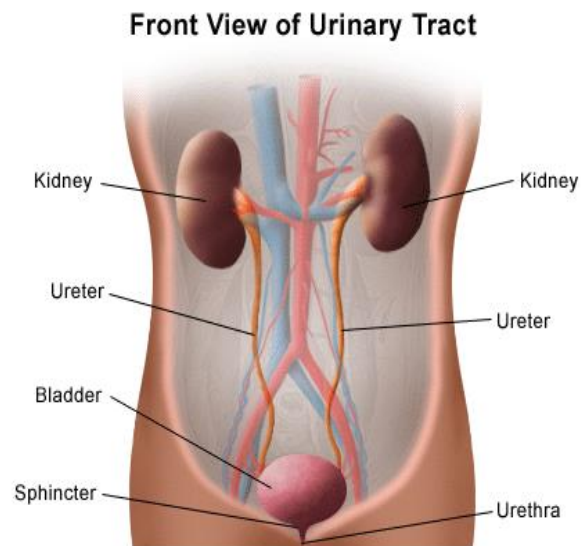
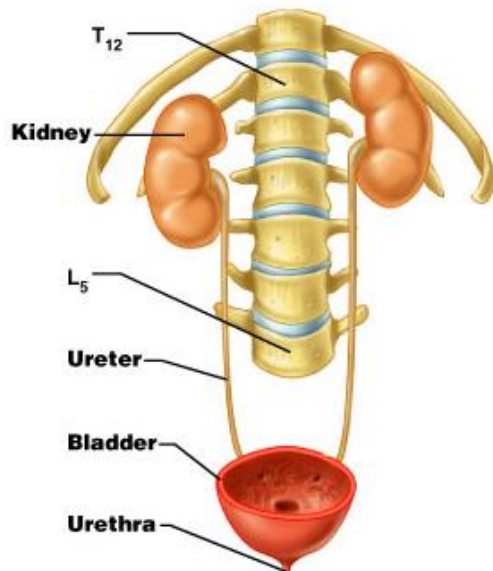
Urinary System



(a)

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image via pinterest.com



Main Functions of the Urinary System

- Excrete waste products from the body.
- Maintain the body's water balance.
- Maintain the body's salt (electrolyte) balance.
- Regulates the blood's acid-base (pH) balance.
- Regulates the body's blood volume.
- Helps to regulate blood pressure (by releasing the enzyme renin)
- Helps stimulate red blood cell production (by releasing the hormone erythropoietin)
- Converts Vitamin D to its active form of calcitriol.
- Dispose of nitrogenous wastes from blood (Urea, Uric acid, and Creatinine)

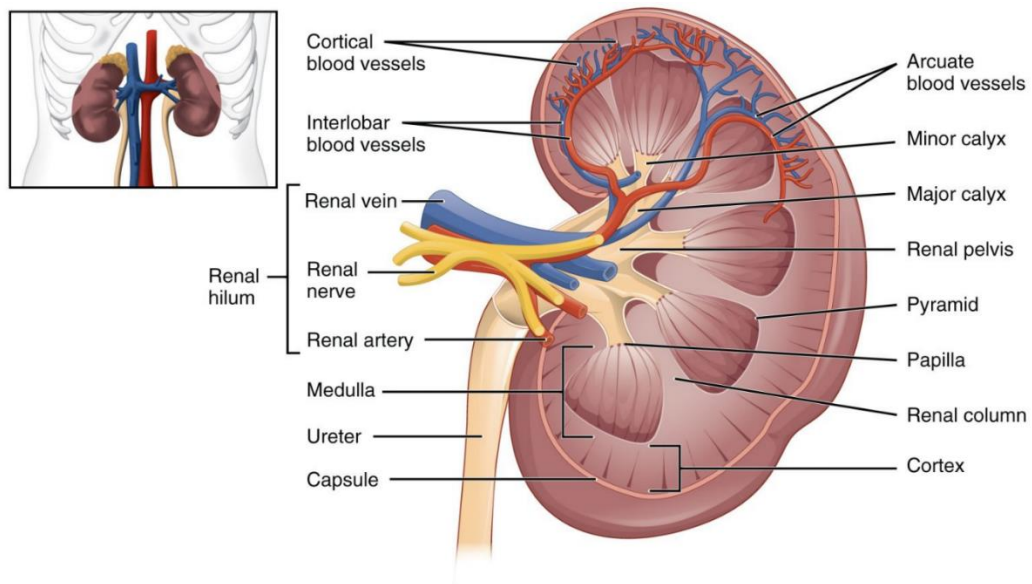
Kidney Structure

The kidneys lie in the retroperitoneal space between the parietal peritoneum and the posterior abdominal wall. Both the right and left kidneys are protected by muscles, fat, and ribs. Kidneys are roughly the size of your fist and are about 11-14 cm in length, 6 cm wide, and 4 cm thick. Kidneys are well vascularized, receiving about 25 percent of the cardiac output at rest.

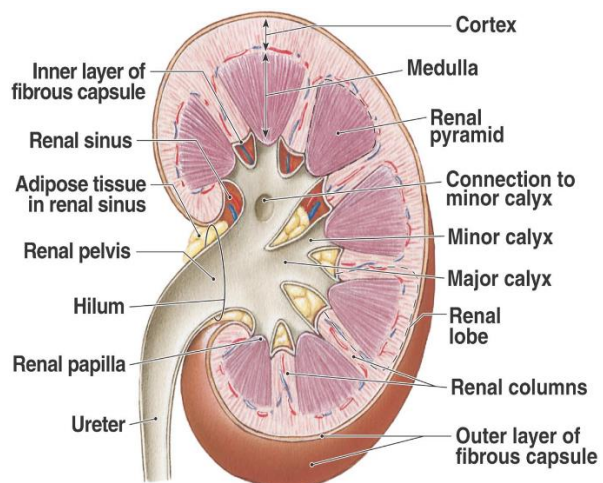
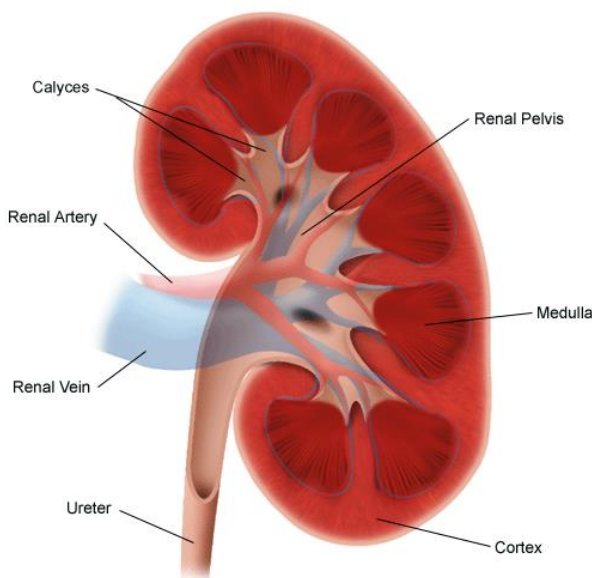
The kidneys, which maintain the purity and constancy of our internal fluids, are perfect examples of homeostatic organs. The two kidneys are reddish organs resembling beans in shape on either side of the abdomen just above the waist and towards the back of the body.

The kidneys contain microscopic filtering units that remove waste, unwanted minerals, and excess water from the blood as urine. Each kidney is connected to the bladder by a long tube called a ureter, which transports urine away.

- Kidneys are retroperitoneal organs (see next slide)
 - Superior lumbar region of the posterior abdominal wall
- Lateral surface is convex
- Medial surface is concave
 - Hilus* is cleft: vessels, ureters and nerves enter and leave
- Adrenal glands* lie superior to each kidney (the yellow blob in pic)



Anatomy of the Kidney



(a) Frontal section of left kidney, anterior view

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The kidney has two regions

Cortex: outer

- Columns of the cortex divide the medulla into “pyramids”

Medulla: inner

- Darker, cone-shaped medullary or renal pyramids
- Parallel bundles of urine-collecting tubules

The human kidney has lobes

- Pyramid and cortical tissue surrounding it
- 5-11 per kidney

Renal pelvis (basin)

- Expanded, funnel-shaped, superior part of the ureter
- Branches to form two or three major calices (seen best on the right pic below)
- Each of these divides again, minor calices: collect urine from papillae of pyramids

Nephrons

Urineriferous tubule (anatomical unit for forming urine)

Nephron

- **Renal corpuscle** (in cortex)
 - Glomerulus (tuft of capillaries)
 - Glomerular (Bowman’s) capsule
- **Tubular section**
 - Proximal convoluted tubule
 - Loop of Henle
 - Distal convoluted tubule
- **Collecting duct**

There are 3 main steps of urine formation:

Glomerular Filtration

- I. Fluid is squeezed out of the glomerular capillary bed

Resorption

- II. Most nutrients, water and essential ions are returned to the blood of the peritubular capillaries

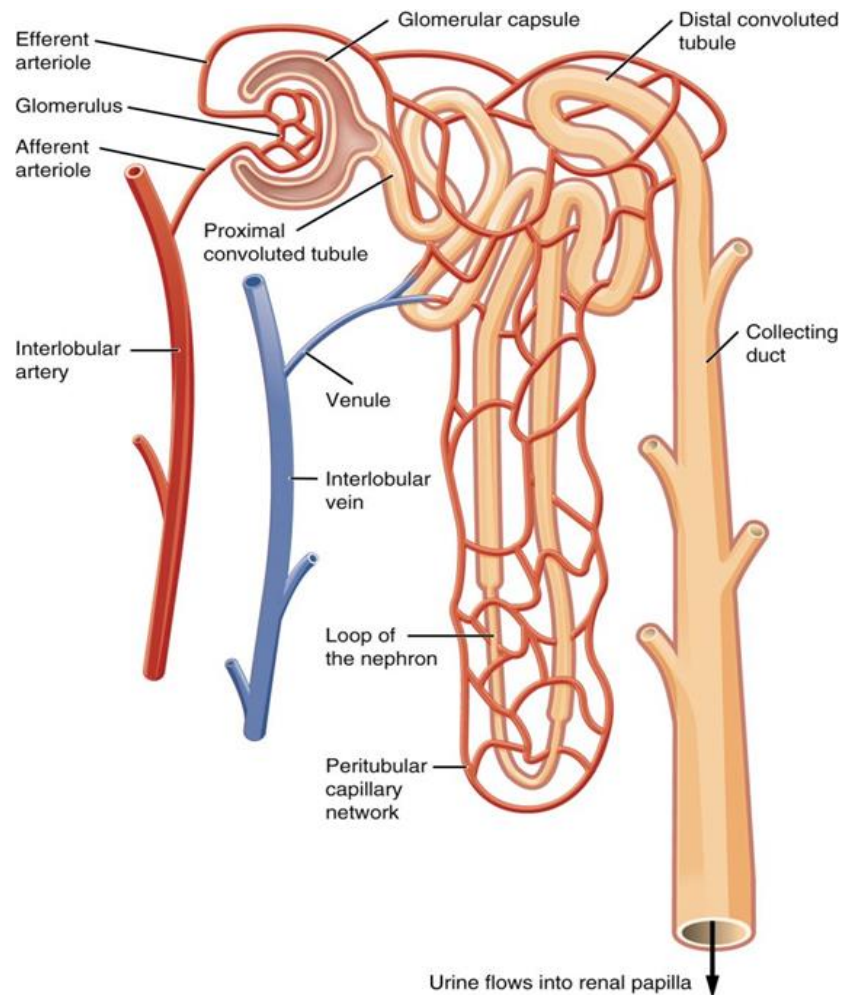
Secretion

- III. Moves additional undesirable molecules into tubule from the blood of peritubular capillaries

These processes ensure that only waste and excess water are removed from the body.

The nephrons are located partly in the cortex and partly inside the renal pyramids, where the nephron tubules make up most of the pyramid mass.

- Nephrons perform the primary function of the kidneys: regulating the concentration of water and other substances in the body.
- They filter the blood, reabsorb what the body needs, and excrete the rest as urine.



Each nephron has a glomerulus, the site of blood filtration.

- The glomerulus is a network of capillaries surrounded by a cuplike structure, the glomerular capsule (or Bowman's capsule).
- As blood flows through the glomerulus, blood pressure pushes water and solutes from the capillaries into the capsule through a filtration membrane.
- This glomerular filtration begins the urine formation process. Ureters

Proximal convoluted tubule.

- This is the part of the tubule that is near to the glomerular capsule.

Loop of Henle.

- The loop of Henle is the hairpin loop following the proximal convoluted tubule.

Distal convoluted tubule.

- After the loop of Henle, the tubule continues to coil and twist before the collecting duct, and this part is called the distal convoluted tubule.

Collecting duct.

- As the tubule extends from the glomerular capsule, it coils and twists before forming a hairpin loop and then again becomes coiled and twisted before entering a collecting tubule called the collecting duct, which receives urine from many nephrons

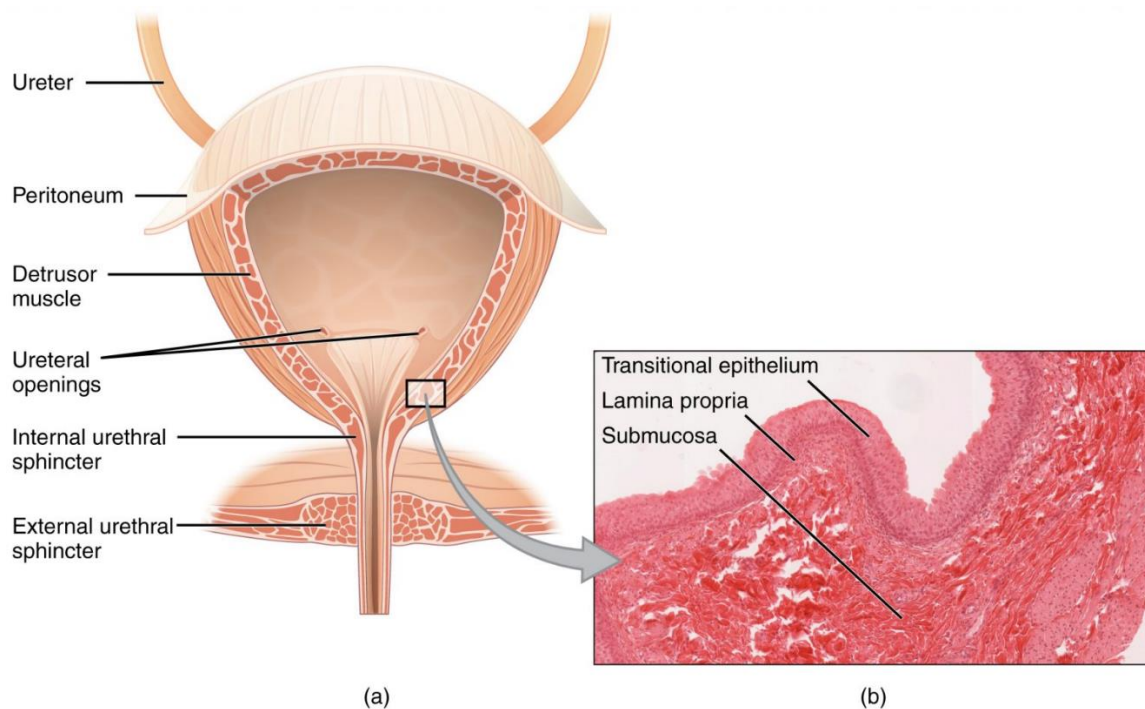
The Ureters

These narrow tubes carry urine from the kidneys to the bladder. Muscles in the ureter walls continually tighten and relax forcing urine downward, away from the kidneys.

Size. The ureters are two slender tubes each 25 to 30 cm (10 to 12 inches) long and 6 mm (1/4 inch) in diameter.

Location. Each ureter runs behind the peritoneum from the renal hilum to the posterior aspect of the bladder, which it enters at a slight angle.

Function. Essentially, the ureters are passageways that carry urine from the kidneys to the bladder through contraction of the smooth muscle layers in their walls that propel urine into the bladder by peristalsis and is prevented from flowing back by small valve-like folds of bladder mucosa that flap over the ureter openings.



Urinary Bladder

The urinary bladder is a smooth, collapsible, muscular sac that stores urine temporarily.

Your bladder holds pee until you're ready to empty it (pee). It's hollow, made of muscle, and shaped like a triangular balloon. Your bladder expands as it fills up. Volumes in adults can range

from nearly zero to 500–600 mL. Peeing is when you empty your bladder. The wall of the bladder also contains irregular crisscrossing bands of smooth muscle called the detrusor muscle

Location. It is located retroperitoneally in the pelvis just posterior to the symphysis pubis.

Function. The detrusor muscles and the transitional epithelium both make the bladder uniquely suited for its function of urine storage.

Urethra

The urethra transports urine from the bladder to the outside of the body for disposal. Just like the rest of the urinary tract organs, much of the urethra is lined by transitional epithelium, except at the end where it is lined by a more protective nonkeratinized stratified squamous epithelium. The internal urethral sphincter is located at the opening into the urethra from the urinary bladder. The internal urinary sphincter consists of smooth muscle that is regulated by an involuntary autonomic nervous system. The external urethral sphincter consists of voluntary skeletal muscle and is located inferior to the internal urinary sphincter.

Internal urethral sphincter. At the bladder-urethral junction, a thickening of the smooth muscle forms the internal urethral sphincter, an involuntary sphincter that keeps the urethra closed when the urine is not being passed.

External urethral sphincter. A second sphincter, the external urethral sphincter, is fashioned by skeletal muscle as the urethra passes through the pelvic floor and is voluntarily controlled.

Female urethra. The female urethra is about 3 to 4 cm (1 1/2 inches) long, and its external orifice, or opening, lies anteriorly to the vaginal opening.

Male urethra. In me, the urethra is approximately 20 cm (8 inches) long and has three named regions: the prostatic, membranous, and spongy (penile) urethrae; it opens at the tip of the penis after travelling down its length.